

FRM 核心考点速通笔记

FOCO 备考速通系列

说明

FRM 考试的主要参考题目是 3 套 PE 以及机经回忆，基于这些题目，我们整理了相应的考点和保姆级的习题详解，希望这份笔记能够帮助大家快速掌握 FRM 考试的核心考点。这份笔记特点如下：

- (1) 在教材整体框架下梳理考点
- (2) 覆盖 PE1、PE2 和 PE3 的 300 道题的考点
- (3) 同时也覆盖近两年的考经考点
- (4) 每个习题都有详细的解析，尽可能做到保姆级地讲解
- (5) 高度浓缩，全是重点
- (6) 单栏排版，80g 护眼纸
- (7) 附有考点目录和 PE 习题索引目录，查找方便
- (8) 可以补充相关要点在上面，形成自己的备考笔记

FRM 备考建议：

1. 聚焦高质量的习题，并结合考经和机经进行复习
2. 注重知识点的理解，而不是死记硬背

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Risk Management and Investment Management

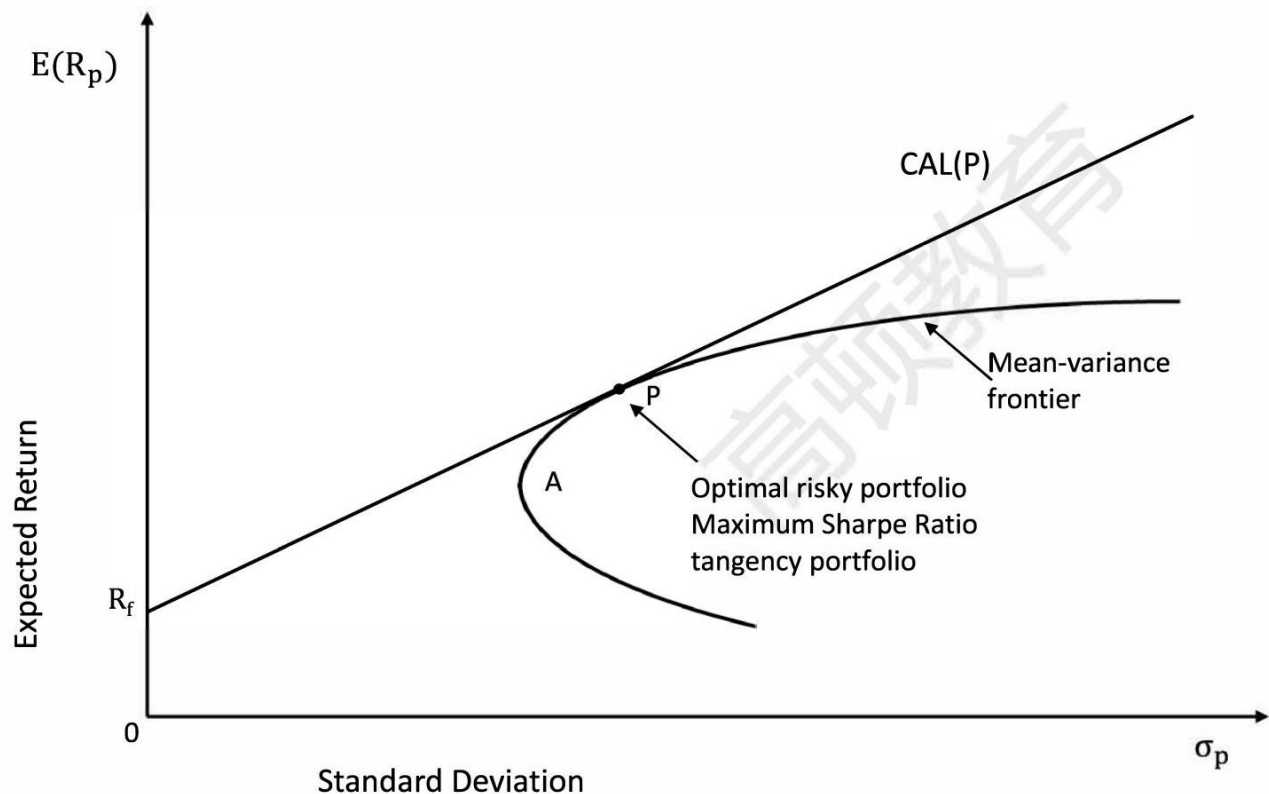
风险管理和投资管理

1 Factor Theory

考点 1 Factor Theory 因子理论

1 Factor theory

- Factors are to assets what nutrients are to food.
 - Factors matter, not assets.
 - * Need to look through asset class labels to understand the factor content.
 - Assets are bundles of factors.
 - * Risk premiums reflect their underlying factor risks.
 - Different investors need different risk factors.
 - * Different investors have different optimal exposures to different sets of risk factors.
- Bad times: Each different factor defines a different set of bad times and the investors who bear these factor risks need to be compensated in equilibrium by earning factor risk premiums.
- Risk premium: Investors are paid risk premiums in normal time as compensation for taking risks.
- Exposure: Assets are bundles of factor risks, and it is the exposures to the underlying factor risks that earn risk premiums.



2 CAPM

- An important theory of factor risk is the CAPM.
 - One factor: market factor. All risky assets have risk premiums determined only by their exposure to the market portfolio.

$$E(R_i) = R_f + \beta(R_m - R_f)$$

- Assumptions:
 - Investors are expected to make decisions solely in terms of expected values and standard deviations of the returns on their portfolios.
 - Investor plan for the same single holding period.
 - Investor have homogeneous expectations or beliefs.
 - An individual cannot affect the asset price by buying or selling action (price taker).
 - All assets, including human capital, are tradable.
 - Markets are frictionless, including no transaction cost and no taxes.
 - All investments are infinitely divisible.
 - Short sale is allowed unlimitedly.
 - Unlimited lending and borrowing at the riskless rate.
- CAPM was revolutionary: the risk of an asset was not how the asset behaved in isolation but how that asset moved in relation(β) to other asset and to the market as a whole.
- CAPM's one failure: the asset risk premiums depend only on the asset's beta and there is only one factor (market factor) matters.
- Lesson 1: Factor - Don't hold an individual asset, hold the factor.
 - Individual stocks: are exposed to the market factor, which carries the risk premium, but also have idiosyncratic risk, which is not rewarded by a risk premium.
 - Market factor portfolio: investors can diversify away the idiosyncratic part and increase their returns.
 - Systematic risk: The market factor is systematic and affects all assets.
- Lesson 2: Optimal exposure - Each investor has his or her own optimal exposure of factor risk
 - Investors have different proportions of the risk-free asset and the market portfolio.
- Lesson 3: Risk premium - The factor risk premium is determined by risk aversion of average investor and the volatility of market.

$$E(R_m) - R_f = \bar{\gamma} \sigma_m^2$$

- $\bar{\gamma}$ is the risk aversion of average investor: 100% in the mean-variance efficient (MVE) portfolio position is the risk aversion of the market.

$$Utility = E(R) - \frac{1}{2} \gamma \sigma^2$$

- Lesson 4: Beta - Risk is factor exposure

$$E(R) - R_f = \beta_i (E(R_m) - R_f)$$

$$\beta_i = \left(\frac{cov(R_i, R_m)}{var(R_m)} \right) = \rho_{i,m} \frac{\sigma_i}{\sigma_m}$$

- The β is a measure of how that stock co-moves with the market portfolio, and the higher the co-movement, the higher the asset's beta.

- Higher betas mean low diversification benefits.
- If the asset pays off when the market has losses, the asset has a low beta. If the payoff of an asset tends to be high in bad times, this is a valuable asset to hold and its risk premium is low.

3 Multifactor models

- Multifactor models recognize the bad times can be defined more broadly than just bad returns on the market portfolio.
 - In equilibrium, investors must be compensated for bearing these multiple sources of factor risk.
- Pricing kernel (stochastic discount factor, SDF)
 - We denote the SDF as m which is an index of bad time. Any risk premium exists because it is compensation for losing money during bad times.

$$m = a + b_1 f_1 + b_2 f_2 + \cdots + b_k f_k$$

$$E(R_i) - R_f = \left(\frac{\text{cov}(R_i, m)}{\text{var}(m)} \right) \left(-\frac{\text{var}(m)}{E(m)} \right) = \beta_{i,m} \lambda_m$$

- Multi-beta relation for an asset's premium:

$$E(R_i) = R_f + \beta_{i,1} E(f_1) + \beta_{i,2} E(f_2) + \cdots + \beta_{i,k} E(f_k)$$

$E(f_k)$ is the risk premium of factor k .

B_{ik} is the beta of asset i with respect to factor k .

4 CAPM (Cont.)

- Failures of the CAPM:
 - CAPM assumes that investors have only financial wealth.
 - * Investors have unique income streams and liabilities.
 - Investors have mean-variance utility.
 - * Asymmetric treatment of risk (more distressed by losses than pleased by gains)
 - Single-period investment horizon.
 - Investors have homogeneous expectations.
 - * In reality, investors do not all share the same beliefs.
 - No taxes or transaction costs.
 - Individual investors are price takers.
 - * The informed investor is trading and moving prices.
 - Information is costless and available to all investors.

5 The fall of efficient market theory

- There are three forms of efficient markets:

Weak form	Asset prices fully reflect all market data.
Semi-strong form	Asset prices reflect all publicly known and available information.
Strong form	Asset prices fully reflect all information , which includes both public and private information.

- Speculative trading is costly, active management is a loser's game and investors cannot beat the market
- Market near-efficiency (Sanford Grossman & Joseph Stiglitz):
 - Assume it is costly to collect information and to trade on that information
 - Active managers earn excess returns as a reward for gathering and acting on costly information.
 - Active managers search for pockets of inefficiency, and in doing so cause the market to be almost efficient.
- The deviations from efficiency have two forms:
 - Rational explanation: high returns compensate for losses during bad times. The key is defining those bad times.
 - Behavioral explanation:
 - * Under- or overreaction to news or events (inefficient updating of beliefs or ignoring some information).
 - * Structural barriers: e.g. entry of capital, regulatory requirements

考点 2 Factors

1 Types of factors

- Assets are buffeted by risk factors. The risk factors offer premiums to compensate investors for bearing losses during bad times.
 - Macro, fundamental-based factors
 - Investment factors
 - * Static factors: simply go long to collect a risk premium (e.g., market factors).
 - * Dynamic factors: can only be exploited through constantly trading different types of securities.

2 Macro factors

- Macro factors affect all investors and the prices of assets. Although macro factors may have different impact on different investors, in general, bad outcomes of macro factors define bad times for the average investor.
- A shock to a factor matters more than the level of the factor.
 - Economic growth
 - Inflation
 - Volatility
- Economic growth:

table 1: Asset Returns and Volatility in Different Economic Conditions

		Large stocks	Small stocks	Govt. bonds	Invest. grade	High yield
Recession	Return	5.6%	7.8%	12.3%	12.6%	7.4%
	volatility	23.7%	33.8%	15.5%	16.6%	18.1%
Expansion	Return	12.4%	16.8%	5.9%	6.0%	7.7%
	volatility	14%	21.2%	9.3%	7.8%	6.8%
Average return		11.3%	15.3%	7%	7%	7.6%

NBER, 1952:Q1 to 2011:Q4 quarterly frequency

- Volatility tends to be very high during bad times. All asset returns are much more volatile during recessions.
- Risky assets generally perform poorly and are much more volatile during periods of low economic growth.

Price for low exposure to growth risk

- Inflation:

table 2: Investor - Investment - Performance Relationship

Investor	Investment	Performance
Can weather recessions	More risky assets	Higher returns in long term to compensate losses during recession
Cannot weather recessions	Less risky assets	Lower returns in long term due to better performance in recession

		Large stocks	Small stocks	Govt. bonds	Investment grade	High yield
Inflation	Low	14.7%	17.6%	8.6%	8.8%	9.2%
	High	8%	13%	5.4%	5.3%	6%
Average return		11.3%	15.3%	7%	7%	7.6%

– ✓ High inflation tends to be bad for both stocks and bonds.

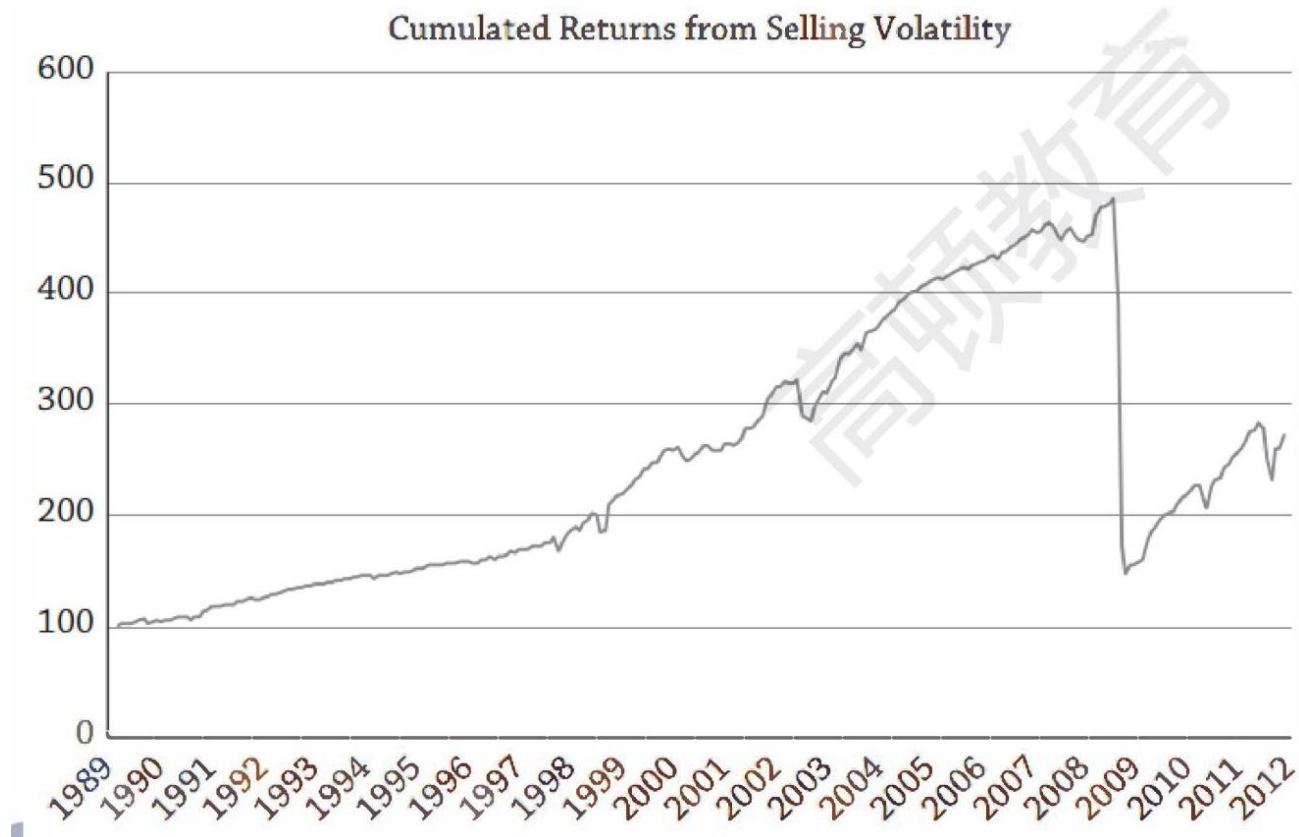
- Volatility:

- Volatility and stock returns: negative relations (leverage effect).
- Bonds offer some respite during periods of high volatility.
- Volatility is negatively linked to the returns of many assets and strategies.
- Volatility is a factor with a negative price of risk
 - * To collect a volatility premium requires selling volatility protection.

- Manage volatility:

- Dislike volatility:
 - * Investors who is sensitive to volatility could increase bond position (but not always pay off in high volatility).
 - * Buy volatility protection (long put, volatility swap)
- Can afford volatility: sell volatility protection (short put)
 - * Sell volatility generates high and steady payoffs during stable times which compensate large losses in a huge crash (inevitable large losses that occur every decade).

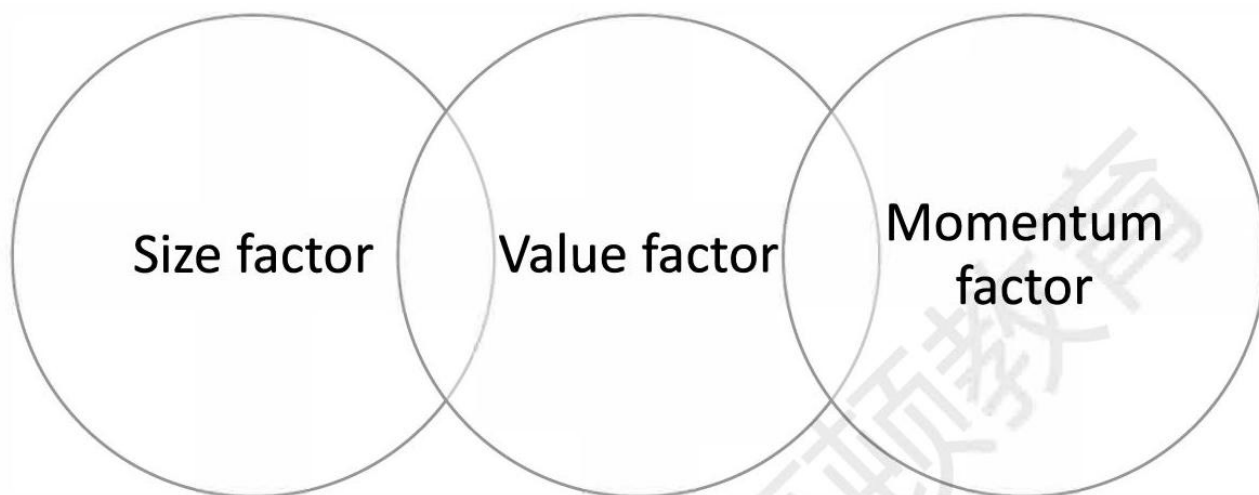
	VIX changes	Stock returns	Bond returns
VIX changes	1	-.39	.12
Stock returns	-.39	1	-.01
Bond returns	.12	-.01	1



- Other macro factors:
 - Productivity risk
 - Demographic risk
 - Political risk

3 Dynamic factors

- Macro factors like inflation and economic growth cannot be directly traded.
- Dynamic factors have a big advantage that they can be easily traded in investors' portfolios □ long-short portfolio
 - Style factors, investment factors, dynamic factors
 - Smart beta, alternative beta
 - Long-short factor



- Fama-French three-factor model: incorporates the systematic factors of market index, firm size (market capitalization) and book-to-market ratio:

$$E(R_i) = R_f + \beta_{i,MKT}E(R_m - R_f) + \beta_{i,SMB}E(SMB) + \beta_{i,HML}E(HML)$$

4 Size factors

- Size factor: The SMB factor was designed to capture the outperformance of small firms relative to large firms.
 - Since the mid-1980s there has not been any significant size effect.
 - Responses to the disappearance of the size effect:
 - * The original discovery of the size premium could have just been data mining.
 - * Rational investors bid up the price of small cap stocks until the effect was removed.

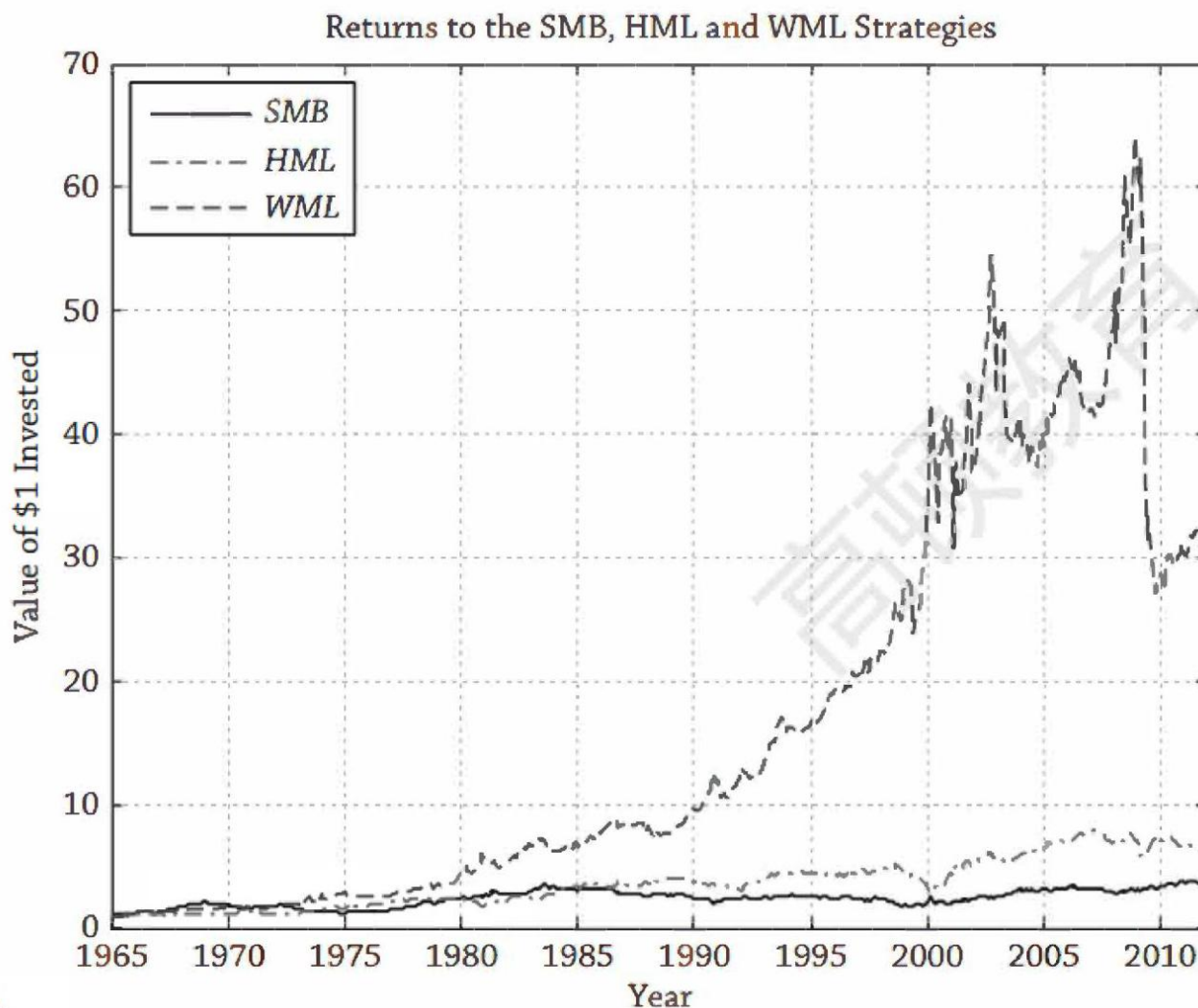
5 Value factors

- Value investing: Long value stocks and short growth stocks.
 - Growth stocks: low book-to-market ratios
 - Value stocks: high book-to-market ratios
- Rational theories of the value premium
 - Flexibility of response to shock: value firms and growth firms differ in how flexible they are and how quickly they can respond to shocks.
 - Value firms are riskier because they cannot shift their firm activities to more profitable activities in bad times (high and asymmetric adjustment costs).
- Behavioral theories of the value premium
 - Overreaction or overextrapolation: investors tend to overextrapolate past growth rates into the future.
 - * Value stocks are cheap because investors underestimate their growth prospects.
 - * Growth firms are expensive because investors overestimate their growth prospects.

6 Momentum factors

- Momentum strategy: is to buy stocks that have gone up over the past six (or so) months (winners) and short stocks with the lowest returns over the same period (loser). Momentum is also called "trend" investing.
 - Momentum effect: winner stocks continue to win and losers continue to lose.
- Four-factor model (Carhart model):

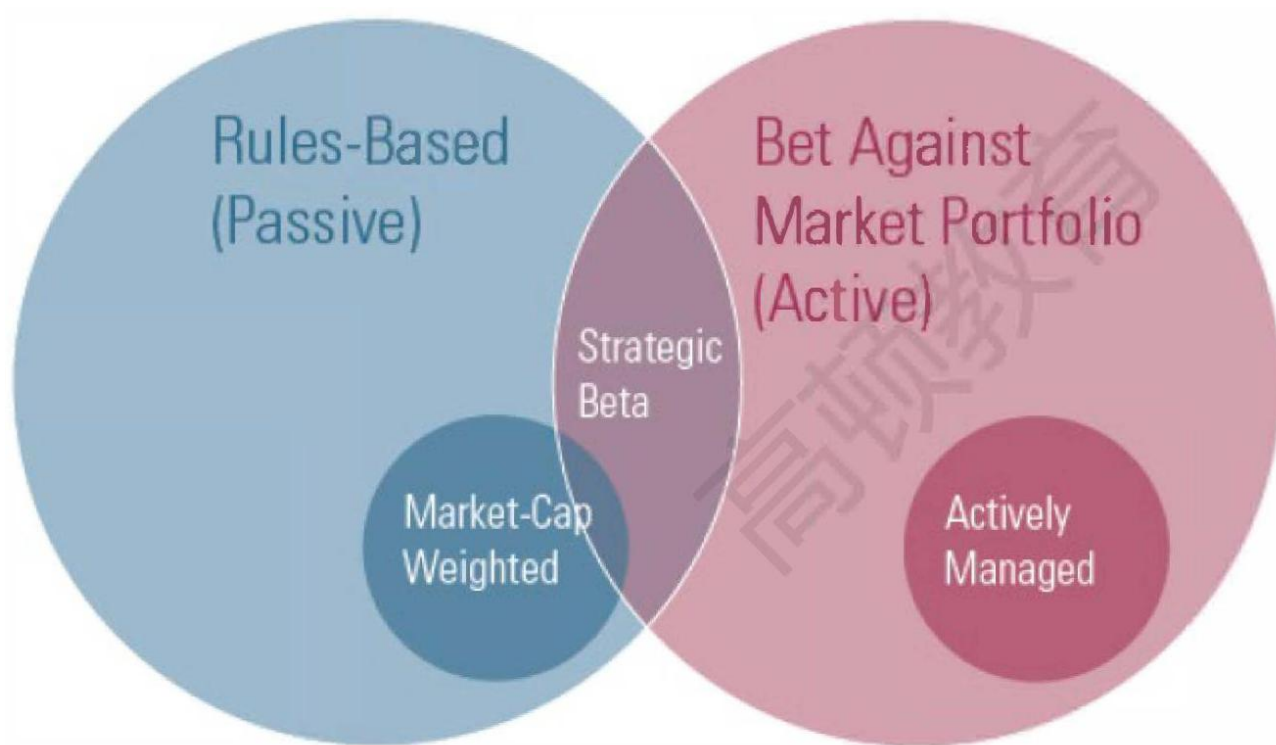
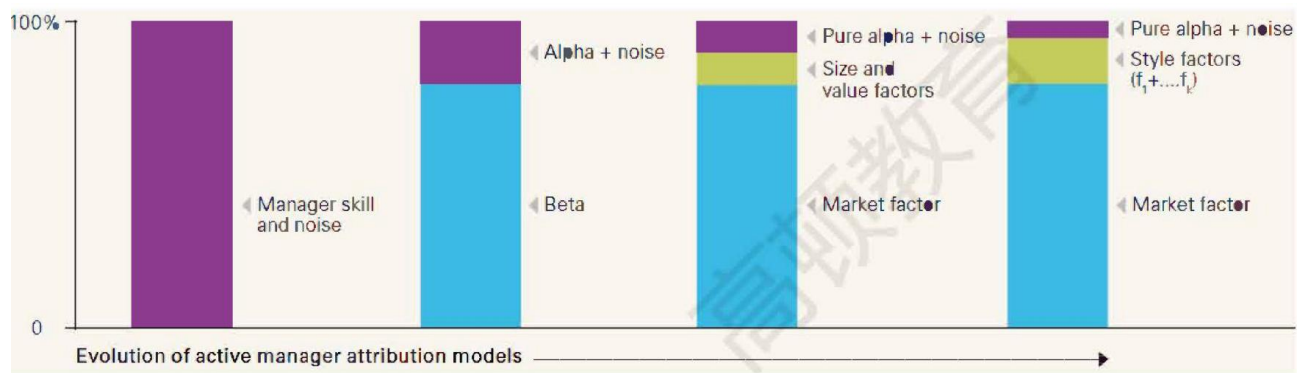
$$E(R_i) = R_f + \beta_{i,MKT}E(R_m - R_f) + \beta_{i,SMB}E(SMB) + \beta_{i,HML}E(HML) + \beta_{i,WML}E(WML)$$

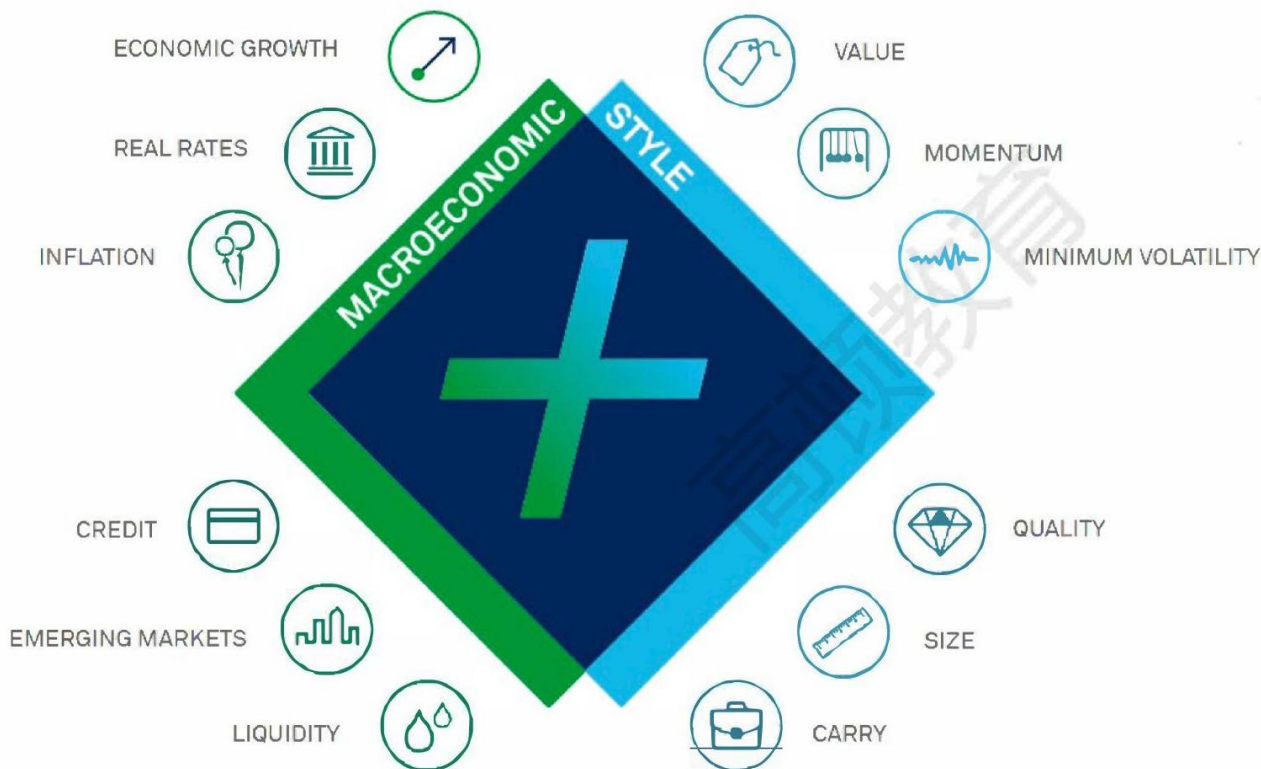


- Characterizing momentum risk:
 - Monetary policy and government risk during extraordinary times: for the loser stocks, government bailouts put a floor underneath the prices of these stocks to stop the tendency to keep losing.
- Behavioral theories of the momentum premium: momentum can be generated in two ways
 - Delayed overreaction: causes the price to persistently drift upward
 - Underreaction: the price does not go up as much as it should have to fully reflect how good the news actually was. Investors then learn and cause the stock to go up again the next period.

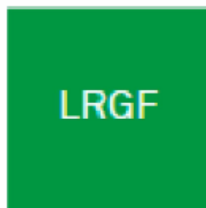
7 Summaries about dynamic factors

- The size, value and momentum strategies are all crosssectional strategy, meaning that it compares one group of stocks against another group of stocks in the cross section.
 - Momentum strategy: positive feedback strategy. Stocks with high past returns are attractive, momentum investors continue buying them and they continue to go up.
 - Value strategy: negative feedback strategy. Stocks with declining prices eventually fall far enough that they become value stocks.





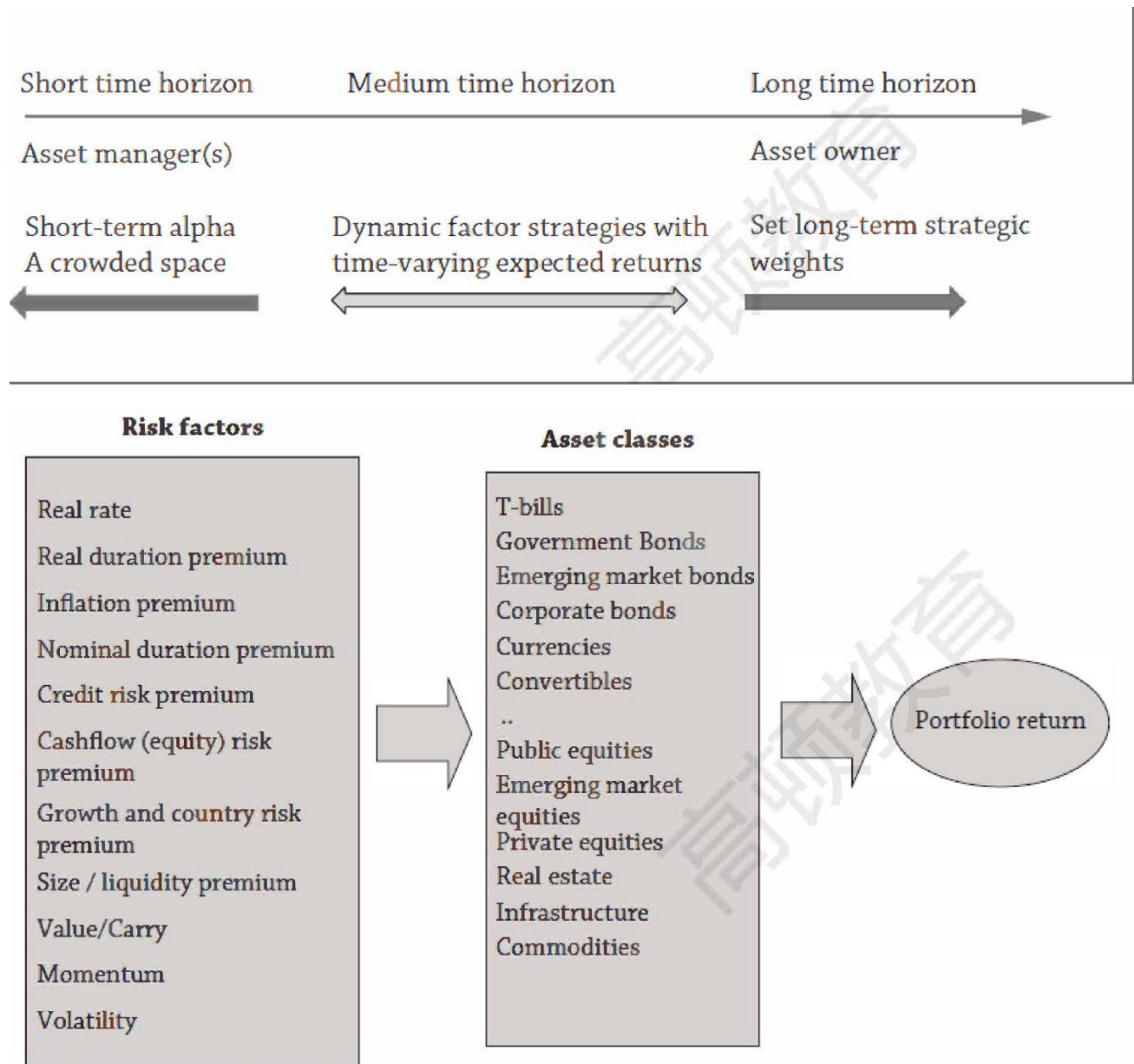
iShares Edge MSCI Min Vol USA ETF
Equity



iShares Edge MSCI Multifactor USA ETF
Equity



iShares Edge MSCI USA Value Factor ETF
Equity



考点 3 Alpha (and the Low-Risk Anomaly)

1 Active management

- Alpha: average return in excess of a benchmark. Benchmark is passive and can be produced without any particular investment knowledge or even human intervention.
 - Excess return (active returns): $r_t^{ex} = r_t - r_t^{bmk}$
 - Average excess return: $\alpha = \frac{1}{T} \sum_{t=1}^T r_t^{ex}$
 - (where there are T observations in the sample)
- Tracking error = $\bar{\sigma} = \text{stdev}(r_t^{ex})$
 - Tracking error constraints are imposed to ensure a manager does not stray too far from the benchmark. The larger the tracking error, the more freedom the manager has.
- Information ratio: the average excess return per unit of risk.

$$IR = \frac{\text{active return}}{\text{tracking error}} = \frac{\alpha}{\bar{\sigma}}$$

- Special case: when the benchmark is risk-free rate

$$\alpha = \overline{r_t - r_{ft}}$$

$$IR = \text{Sharpe Ratio} = \frac{\overline{r_t - r_{ft}}}{\sigma}$$

- Grinold fundamental law:

$$IR \approx IC \times \sqrt{BR}$$

- Information coefficient (IC): the correlation of the manager's forecast with the actual returns
- The breadth of the strategy (BR): how many bets are taken
- Creating alpha: by making bets that deviate from that benchmark. The key to generating alpha is forecasting.
 - How good asset managers at forecasting
 - How many bets
- Assumptions of fundamental law: each active return is independent from the other active return forecasts for that period and independent from the forecast for that security in subsequent periods.
- Limitations of fundamental law :
 - ICs are assumed to be constant across BR.
 - Manager whom you find may truly have a high IC, but the one hundredth manager whom you hire probably does not.
 - It is difficult to have truly independent forecasts in BR.

2 Benchmarks

- Ideal benchmark:
 - Well defined
 - Tradable
 - Replicable
 - Adjusted for risk
- Index benchmark

$$r_t - r_{ft} = 0.0344 + 0.7272 \times (r_t^{R1000} - r_{ft}) + \varepsilon_t \quad (1)$$

$$r_t = \underbrace{0.0344}_{\downarrow \alpha} + \underbrace{0.2728 r_{ft} + 0.7272 r_t^{R1000}}_{\downarrow r_{bmk}} + \varepsilon_t \quad (2)$$

- ✓ The benchmark portfolio holding 27% in the risk - free asset and 73% in the Russell 100
- ✓ Naïve benchmark

$$r_t = \underbrace{0.015}_{\downarrow \alpha} + \underbrace{r_t^{R1000}}_{\downarrow r_{bmk}} + \varepsilon_t \quad (3)$$

- Factor benchmarks are a combination of factors

- The factor benchmark describes the systematic components of asset i 's return.

$$E(r_i) - r_f = \beta_i [E(r_m) - r_f]$$

$$\text{if } \beta_i = 1.3 \rightarrow E(r_i) = -0.3r_f + 1.3E(r_m)$$

$$E(r_i) = \alpha_i + \boxed{[-0.3r_f + 1.3E(r_m)]} \downarrow_{E(r_{\text{bnk}})}$$

- Factor regressions

$$r_{it} - r_{ft} = \alpha + \beta(r_{mt} - r_{ft}) + sSMB_t + hHML_t + uUMD_t + \varepsilon_{it}$$

- $s = 0$ if a stock co - moves neither with small nor large stocks, it's a medium - size stock
- $s > 0$ stock moves together with small stocks
- $s < 0$ stock moves together with large stocks

	Coefficient	T - stat
Alpha	0.68%	2.05
MKT loading	0.66	8.26
SMB loading	-0.5 (large stock exposure)	-4.86
HML loading	0.36 (value orientation)	3.33
UMD loading	-0.04	-0.66
Adj R^2	0.27	

- Statistical significance of alpha: determines whether the alpha reliably indicates ability.

- * Null (H_0): Alpha is zero

- * Alternative (H_A): Alpha is not zero

$$t(\hat{\alpha}) = \frac{\hat{\alpha} - 0}{\hat{\sigma}(\varepsilon)/\sqrt{N}}$$

- Time-varying factor: introduced "style analysis" to handle time-varying benchmarks where the factor exposures evolve through time.

- Style analysis seeks to rectify two potential shortcomings of factor regression:

- * The Fama-French portfolios are not tradable.

- * The factor loadings may vary over time.

- The collection of index funds that replicate the fund is called the "style weight."

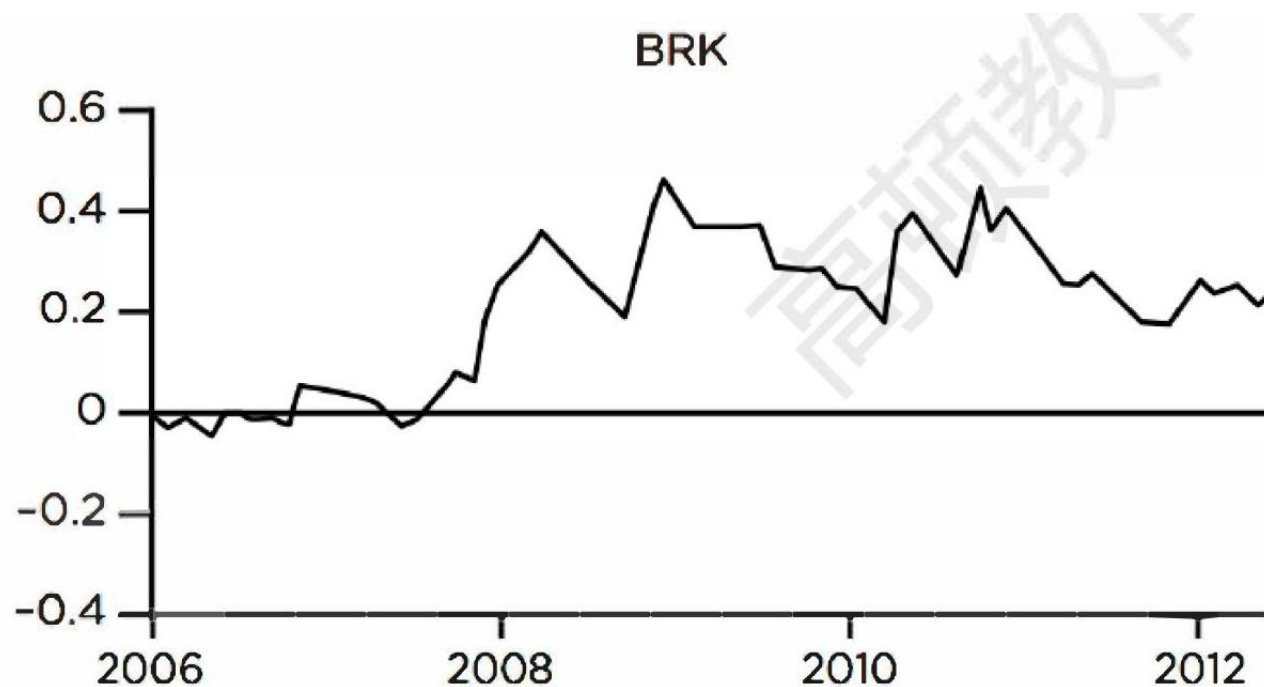
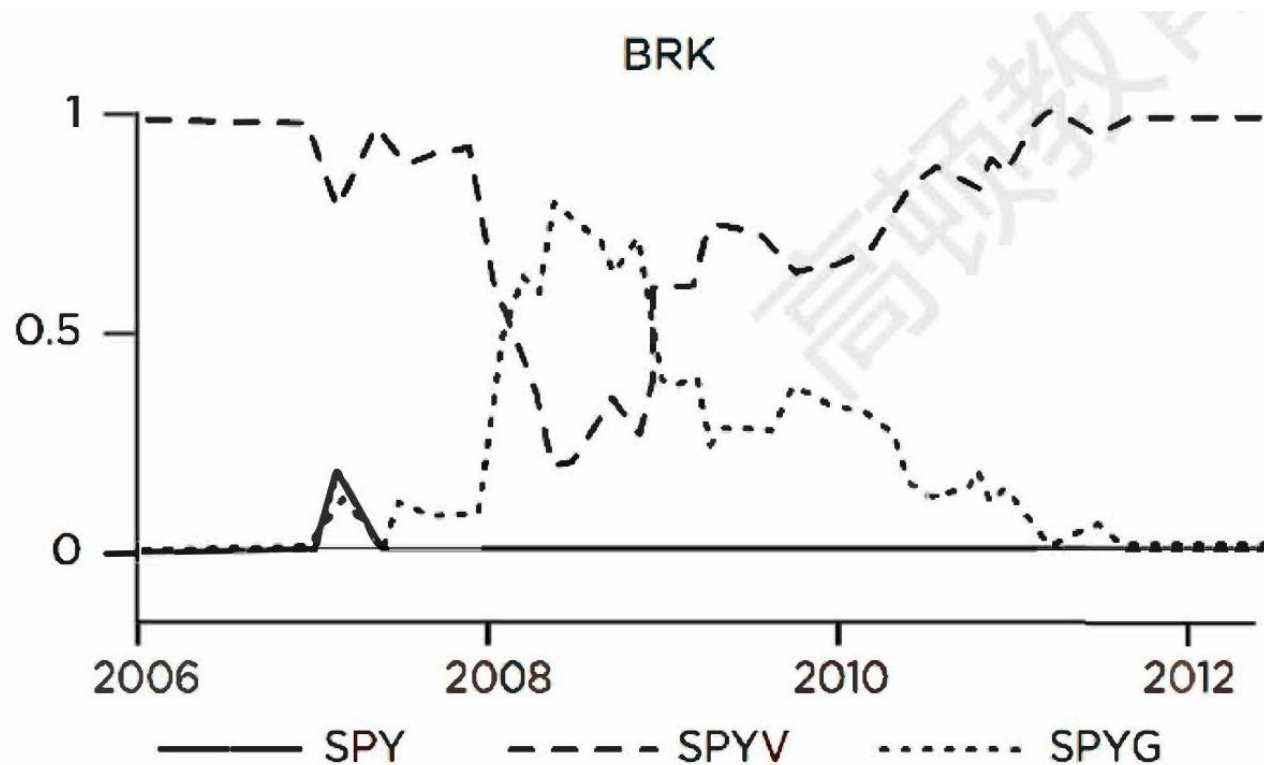
- Time-varying factor

- Example: Style analysis with no shorting - Berkshire Hathaway Inc. (data from Jan. 2001 to December 2011)

$$\begin{aligned} r_{t+1} = & \alpha_t + \beta_{SPY,t}SPY_{t+1} \\ & + \beta_{SPYV,t}SPYV_{t+1} \\ & + \beta_{SPYG,t}SPYG_{t+1} + \varepsilon_{t+1} \end{aligned}$$

- α_t
- $\beta_{SPY,t}SPY_{t+1}$: SPY: SPDR S&P 500 ETF, which is designed to mimic the S&P 500.
- $\beta_{SPYV,t}SPYV_{t+1}$: SPYV: SPDR S&P 500 Value ETF, which tracks the S&P 500 value index.
- $\beta_{SPYG,t}SPYG_{t+1}$: SPYG: SPDR S&P 500 Growth ETF, which replicates the S&P 500 growth index.
- ε_{t+1}

- Time-varying factor
 - * Style analysis with no shorting



Cumulated Excess Returns

- Non-linear payoffs: alphas are computed in a linear framework. There are many nonlinear strategies, especially those involving dynamic option strategies, that can masquerade as alpha.
- Why do dynamic, nonlinear strategies yield false measures of alpha?
 - Buying and selling options changes the distribution of returns.

3 Low-risk anomaly

- Traditional financial theory: in CAPM for example, there should be a positive relation between risk and return.
- The risk anomaly: stocks with low betas and low volatilities have high returns.
 - Volatility is negatively related to future returns.
 - Realized beta is negatively related to future returns.
 - Minimum variance portfolios do better than the market.
- Explanation:
 - Leverage constraints: investors wish to take on more risk but are unable to take on more leverage.
 - Agency problems: Institutional managers have constraints.
 - * Long only: If a stock has negative alpha, manager needs to short stock and he or she cannot do so. The tracking error constraint also limits the underweight position.
 - Preference: Investors have a preference for high-volatility stocks, then they bid up stocks until they have low returns (and vice versa).

2 Portfolio risk Management

考点 1 Portfolio Construction

1 5 inputs to the portfolio construction process

- Current portfolio: near certainty
- Alphas: are often unreasonable and subject to hidden biases
- Covariances
- Active risk aversion: target level of active risk consistent with an active risk aversion
- Transactions costs

2 Refining alphas

- Motivation: Most active managers construct portfolios subject to certain constraints, agreed upon with the client. These limits can make the portfolio less efficient, but they are hard to avoid.
 - Not take short positions.
 - Limit the amount of cash in the portfolio.
 - Liquidity.
- Alphas can be adjusted so that they are in line with investor or manager's desires for risk control and various constraints.
- Scale the alphas

$$\alpha = \text{volatility} \cdot IC \cdot \text{Score}$$

$\uparrow$
 Residual risk

(4)

$$\alpha \in [0 - z \times IC \times \text{Volatility}, 0 + z \times IC \times \text{Volatility}]$$

- Score has a mean of zero and standard deviation of one.
- Information coefficient: Correlation between actual and forecasted outcomes.

- Example: An analyst regresses the returns of 200 stocks against the returns of a major market index. The resulting pool of 200 alphas has a residual risk of 25% and an information coefficient of 8%. If the alphas are normally distributed with a mean of 0%, roughly how many stocks have an alpha greater than 4% or less than -4%?
- Answer: The interval of α is $[0 - 25\% \cdot 8\% \cdot z, 0 + 25\% \cdot 8\% \cdot z]$

$$\rightarrow z = 2$$

The total number of stocks is 200, so there are 10 alphas that are out of the range.

- Trim alpha outliers: very large positive or negative alphas can have undue influence.
 - Closely examine all stocks with alphas greater in magnitude than three times the scale of the alphas.
- Neutralization: If our initial alphas imply an alpha for the benchmark, the neutralization process recenters the alphas to remove the benchmark alpha.
 - Benchmark-and cash-neutral alphas: Adjusting the benchmark alpha to zero. The optimal position that uses benchmark will have a beta of one.
 - Risk-Factor-neutral alphas: will include only information on the factors the manager can forecast.
- Neutralization:
 - Benchmark-neutral alphas:
 - Example 1: shift the alpha of the benchmark Major Market Index from 1.6 basis points to 0.

Stock	Beta	Alpha	Modified Alpha
American Express	1.21	-1.14%	-1.16%
AT&T	0.96	0.30%	0.29%
Chevron	0.46	0.11%	0.10%
Coca Cola	0.96	-0.78%	-0.79%
Disney	1.23	0.60%	0.58%
Dow Chemical	1.13	0.22%	0.20%
DuPont	1.13	-0.65%	-0.67%

table 3: American Express Modified Alpha = -1.14% - 1.21 · 0.00016

- Example 2: You want to construct a portfolio so that all of the alphas are benchmark-neutral. Stock XYZ has a volatility of 30%, an information coefficient of 0.2, and alpha of 90 basis points, and a beta of 1.94. The benchmark has an alpha of 3.7 basis points. What is the appropriate benchmark-neutral alpha for stock XYZ?
- Answer: modified benchmark-neutral alpha = $0.009 - 0.00037 \cdot 1.94 = 0.0083$

3 Active risk aversion

- Determination of risk aversion:
 - Risk aversion(λ_A) = $\frac{IR}{2 \times \sigma_\alpha}$
 - * Utility (Value added) = $\alpha_p - \lambda_A \cdot \sigma_\alpha^2$
- Specific factor risk
 - Decomposition of risk allows differing aversions to these different sources of risks.
 - * Utility = $\alpha_p - (\lambda_{A,CF} \cdot \sigma_{A,CF}^2 + \lambda_{A,SP} \cdot \sigma_{A,SP}^2)$
 - * Since specific risk arises from bets on specific assets, a high aversion to specific risk **reduces bets** on any one stock.
 - * For managers of multiple portfolios, aversion to specific risk can help **reduce dispersion**.

4 Transaction costs

- Transactions costs: are the costs of moving from one portfolio allocation to another.
 - Rebalancing incurs transactions costs at that point in time. Transaction costs must be amortized over the investment horizon in order to determine the optimal portfolio adjustments.
- Revisions and rebalancing: fund manager should make the correct trade-off between expected active return, active risk, and transactions costs.
 - If the manager is not sure of his or her ability to correctly specify the alphas, the active risk, and the transactions costs, less frequent revision may be a safeguard.
 - The shorter time horizon for forecasting alphas, the larger amounts of noise. Rebalancing for very short horizons would involve frequent reactions to noise, not signal.

- Revisions and rebalancing:

$$MCVA_n = \alpha_n - 2 \times \lambda_A \times \sigma_\alpha \times MCAR_n$$

- MCVA (Marginal contribution to value added): shows how value added, as measured by risk-adjusted alpha, changes as the holding of the stock is increases.
- MCAR (Marginal contribution to active risk): measures the rate at which active risk changes as we add more of stock n. The change in value added also depends upon the impact (at the margin) on active risk of adding more of stock n.
- Revisions and rebalancing: In order to make rebalancing decision, we can compare the $MCVA_n$ for stock n to the transactions costs.

$$MCVA_n = \alpha_n - 2 \times \lambda_A \times \sigma_\alpha \times MCAR_n$$

- No trade region:

$$-SC_n \leq MCVA_n \leq PC_n$$

$$2 \times \lambda_A \times \sigma_\alpha \times MCAR_n - SC_n \leq \alpha_n \leq PC_n + 2 \times \lambda_A \times \sigma_\alpha \times MCAR_n$$

- Determination of the no-trade region: transactions costs, risk aversion, alpha and the riskiness of the assets.

5 Portfolio construction techniques

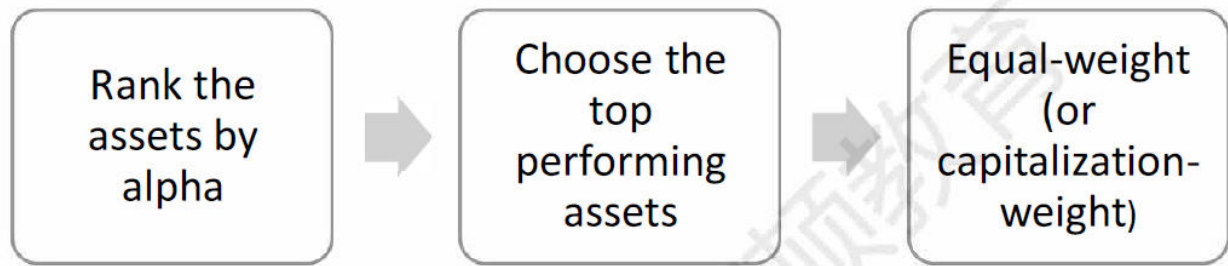
- Objective: maximizing active returns minus an active risk penalty.

$$\text{Value added} = \alpha_p - \lambda_A \cdot \sigma_\alpha^2 - \text{transaction cost}$$

- Screen(筛选)
- Stratification(分层)
- Linear programming(线性规划)
- Quadratic programming(二次规划)

What procedures can we use to make the portfolio construction process robust in the presence of unreasonable and noisy inputs?

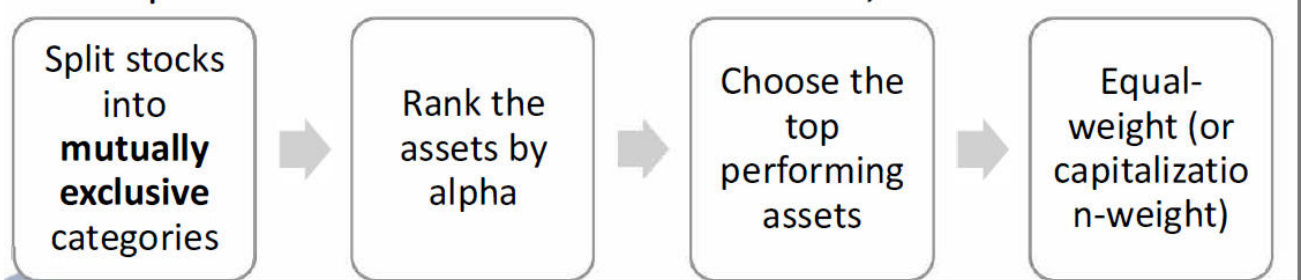
- Screen:



- Divide the universe of assets into buy, hold, and sell decisions based on the rankings.
- Purchase any assets on the buy list not currently in the portfolio, and sell any assets in the portfolio that are on the sell list.
- Advantages:
 - * Simple; easy to understand and computerize
 - * Clear link between cause (buy, sell or hold) and effect (portfolio performance).
 - * The screen is robust. It depends solely on ranking so wild estimates of alphas will not alter the result.
 - * Enhances alphas by concentrating the portfolio in the high-alpha stocks
- Risk control:
 - * Include a sufficient number of stocks and by weighting them to avoid concentration in any single stock.
 - * Transactions costs are limited by controlling turnover.
- Disadvantages:
 - * Ignore all information in the alphas apart from the rankings.
 - * Excluded assets with lower alphas.

• Stratification:

- Split the list of assets into mutually exclusive categories.
- Risk control: make sure that the portfolio has a representative holding in each category. Chosen well, the categories can lead to reasonable risk control while some important risk dimensions are excluded, risk control will fail.



- Linear programming: characterizes assets along dimensions of risk, e.g. industry, size, volatility, and beta (without making the dimensions mutually exclusive).
 - Advantages: takes all the information about alpha into account and controls risk by keeping the characteristics of the portfolio close to the characteristics of the benchmark.
 - Disadvantages:
 - * Hard to produce portfolios with a prespecified number of stocks.
 - * The risk-control characteristics should not work at cross purposes with the alphas.
- It also includes explicit transactions costs, a limit on turnover, and upper and lower position limits on each asset.

- Objective of linear programming: maximize the portfolio's alpha less transactions costs while remaining close to the benchmark portfolio in the risk control dimensions.
- Quadratic programming: explicitly considers alpha, risk, and transactions costs.
 - Advantage: Including all the constraints and limitations one finds in a linear program.
 - Disadvantage: A great many more inputs than the other portfolio construction techniques. More inputs mean more noise.

6 Causes and Methods for Dispersion

- Dispersion: fund managers run separate accounts for multiple clients but the portfolio returns are not identical.
 - Dispersion is the difference between the maximum return and minimum return for separate account portfolios.
- Dispersion:
 - Client-driven: constrains from clients
 - Manager-driven: a lack of attention (separate accounts exhibit different betas and different factor exposures through lack of attention).
 - If transactions costs were zero, dispersion would disappear, at no cost to investors.

考点 2 Portfolio Risk: Analytical Methods

1 Portfolio VaR measures

- Delta-normal model: all individual security returns are assumed normally distributed.
- Traditional portfolio analysis: based on variances and covariances, which is why it is sometimes called the covariance matrix approach.
- Individual VaR: The VaR of one component taken in isolation.
 - Transactions costs: are the costs of moving from one portfolio allocation to another.
 - * Rebalancing incurs transactions costs at that point in time. Transaction costs must be amortized over the investment horizon in order to determine the optimal portfolio adjustments.
 - Revisions and rebalancing: fund manager should make the correct trade - off between expected active return, active risk, and transactions costs.
 - * If the manager is not sure of his or her ability to correctly specify the alphas, the active risk, and the transactions costs, less frequent revision may be a safeguard.
 - * The shorter time horizon for forecasting alphas, the larger amounts of noise. Rebalancing for very short horizons would involve frequent reactions to noise, not signal.
 - Revisions and rebalancing:

$$MCVA_n = \alpha_n - 2 \times \lambda_A \times \sigma_\alpha \times MCAR_n$$

- * MCVA (Marginal contribution to value added): shows how value added, as measured by risk - adjusted alpha, changes as the holding of the stock is increases.
- * MCAR (Marginal contribution to active risk): measures the rate at which active risk changes as we add more of stock n. The change in value added also depends upon the impact (at the margin) on active risk of adding more of stock n.

- Revisions and rebalancing: In order to make rebalancing decision, we can compare the $MCVA_n$ for stock n to the transactions costs.

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* Determination of the no - trade region: transactions costs, risk aversion, alpha and the riskiness of the assets.

- Diversified VaR: takes into account diversification benefits between components.
- VaR for portfolio of two assets:
- Undiversified VaR: is the sum of individual VaRs
- Diversification benefit: the difference between the diversified VaR and the undiversified VaR which typically is shown in VaR reporting systems.
- Example: A risk manager is evaluating a pairs trading strategy recently initiated by one of the firm's traders . The strategy involves establishing a long position in Stock A and a short position in Stock B. The following information is also provided:
1- day 99% VaR of Stock A is USD 100 million
1-day 99% VaR of Stock Bis USD 125 million
The estimated correlation between long positions in Stock A and Stock B is 0.8. Assuming that the returns of Stock A and Stock Bare jointly distributed, the 1-day 99% VaR of the combined positions is closest to?
- Answer:

$$\begin{aligned} VaR_p &= z \cdot V \cdot \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho_{12}} \\ &= \sqrt{VaR_1^2 + VaR_2^2 + 2VaR_1 VaR_2 \rho_{12}} \\ &= \sqrt{100^2 + 125^2 + 2 \cdot (-0.8) \cdot 100 \cdot 125} = 75 \end{aligned}$$

Since this is a pairs trading strategy with a long and a short position, the proper correlation to use is -0.8

- Role of correlation on portfolio risk:

$$\sigma_p = \sigma \sqrt{\frac{1}{N} + \left(1 - \frac{1}{N}\right) \rho}$$

- All assets have the same risk.
- All correlations are the same.
- Equal weight is put on each asset.

2 VaR tools

- Marginal VaR (MVaR): the change in portfolio VaR resulting from taking an additional dollar of exposure to a given component.

$$\begin{aligned} MVaR_i &= \frac{\partial VaR}{\partial(\text{investment})} = z \frac{\partial \sigma_p}{\partial w_i} \\ &= z \frac{\text{cov}(R_i, R_p)}{\sigma_p} = z \cdot \sigma_i \cdot \rho_{ip} = z \cdot \beta_i \cdot \sigma_p = \frac{VaR_p \cdot \beta_i}{V_p} \end{aligned}$$

- Example: There is a portfolio which is made up of two assets, stock ABC and stock XYZ. Assume the portfolio is valued at \$3 million and the 95% VaR of the portfolio is \$257,738. Stock ABC has a beta of 0.6146. Please calculate the Marginal VaR of stock ABC.

- Answer:

$$MVaR_{ABC} = \frac{VaR_p \cdot \beta}{V_p} = \frac{0.6146 \cdot 257,738}{3,000,000} = 0.0528$$

- Incremental VaR: is the change in VaR owing to a new position.

$$\text{Full Incremental VaR} = VaR_{p+a} - VaR_p$$

- The main **drawback** of this approach is that it requires a **full revaluation** of the portfolio VaR with the new trade (quite time - consuming for large portfolios).
- **Incremental VaR**: $\approx MVaR_i \cdot V_i$
- Component VaR: A partition of the portfolio VaR that indicates how much the portfolio VaR would change approximately if the given component was deleted.

$$\begin{aligned} CVaR_i &= MVaR_i \times V_i \\ &= MVaR_i \times w_i \times V_p = VaR_p \times \beta_i \times w_i \end{aligned}$$

Percentage contribution (%) to VaR of component:

$$\begin{aligned} i &= \frac{CVaR_i}{VaR_p} = w_i \beta_i \\ CVaR_1 + CVaR_2 + \dots + CVaR_n &= VaR_p \left(\sum_{i=1}^n w_i \beta_i \right) = VaR_p \end{aligned}$$

- Example 1:

There is a portfolio which is made up of two assets, stock ABC and stock XYZ. Assume the portfolio is valued at \$3 million (Stock ABC \$2million, stock XYZ \$1million) and the 95% VaR of the portfolio is \$257,738. Marginal VaR of stock ABC is 0.0528. Please calculate the Component VaR of stock XYZ.

- Answer:

$$\begin{aligned} CVaR_{ABC} &= MVaR_i \times V_i \\ &= 0.0528 \cdot 2,000,000 = 105,600 \\ CVaR_{XYZ} &= 257,738 - 105,600 = 152,138 \end{aligned}$$

– Example 2:

Please calculate the percent of contribution to VaR of each component.

$$i_{ABC} = \frac{CVaR_{ABC}}{VaR} = \frac{105,600}{257,738} = 40.97\%$$

$$i_{XYZ} = \frac{CVaR_{XYZ}}{VaR} = \frac{152,138}{257,738} = 59.03\%$$

3 Risk management

- Only taking risk into consideration → global minimum

A manager can lower a portfolio VaR by:

- Lowering position with the highest marginal VaR.
- Adding position with the lowest marginal VaR.

- When the portfolio risk has reached a global minimum:

- All $MVaR_i$ must be equal.
- All betas must be equal.

- Taking return and risk into consideration → optimal portfolio

- Maximize the Sharpe ratio with VaR of the portfolio:

$$\frac{\bar{R}_p - \bar{R}_f}{VaR_p}$$

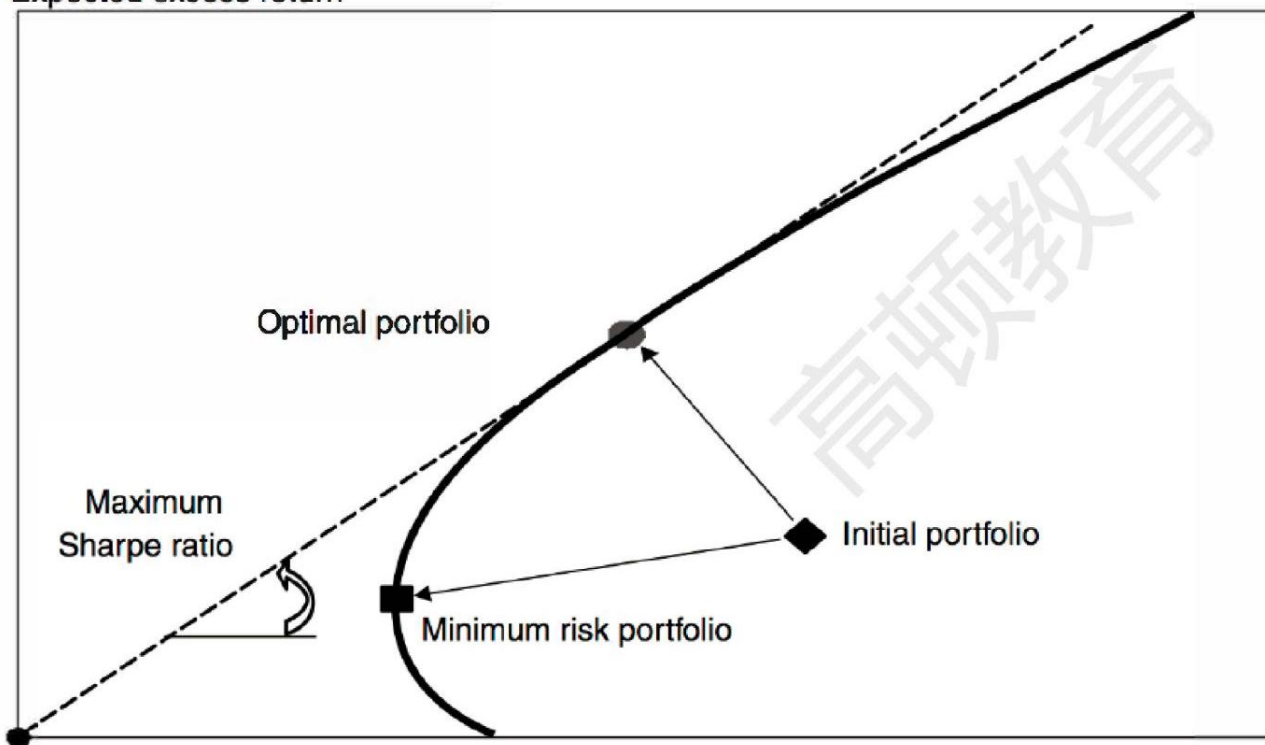
- Lowering allocation to the position with the lowest excess expected return to marginal VaR.
- Adding allocation to the position with the highest excess expected return to marginal VaR.

$$\frac{R_i - R_f}{MVaR_i}$$

- When the portfolio risk has reached a optimal portfolio:

- All the ratios of $\frac{R_i - R_f}{MVaR_i}$ must be equal.
- All the ratios of $\frac{R_i - R_f}{\beta_i}$ must be equal.

Expected excess return



考点 3 VaR and Risk Budgeting in Investment Management

1 Types of risks

- Absolute risk vs. Relative risk
- Policy-mix vs. Active-management
- Funding risk

2 Absolute vs. relative risk

- Investment asset managers can have different perceptions of risk.
- Absolute risk: is the risk of a dollar loss over the horizon.
- Relative risk: is the risk of a dollar loss in a fund relative to its benchmark.
 - If the excess return of the asset over the benchmark is normally distributed, VaR can be measured from the relative perspective: $VaR_{\text{relative}} = z \cdot \sigma_{\alpha} \cdot V$

3 Policy-mix vs. active-management

$$R_{\text{asset}} = R_{\text{policy mix}} + R_{\text{active management}} = \sum_i w_i^b R_i^b + \sum_i (w_i R_i - w_i^b R_i^b)$$

- Policy-mix risk: is the risk of a dollar loss owing to the policy mix selected by the fund. The risk from passive funds investment, this risk represents that of a passive strategy.
- Active-management risk: is the risk of a dollar loss owing to the total deviations from the policy mix.
- Most of the risk is due to the policy mix.

- The active-management VaR is rather small.
- The policy-mix VaR and active-management VaR do not add up to the total-asset VaR.

4 Funding risk

- Funding risk 相关概念: surplus
- Δ surplus
- Surplus at risk
- Confidence interval of surplus
- A pension fund with defined benefits promises a stream of fixed payments to retirees. If the assets are not sufficient to cover these liabilities, the shortfall will have to be made up by the fund's owner (provide additional contributions to the fund).
 - Risk should be viewed in an asset liability management (ALM) framework.
 - Funding risk: the value of assets will not be sufficient to cover the liabilities of the fund.

5 Funding risk - surplus

- Surplus: is the difference between the value of the assets and the liabilities.
 - If liabilities consist mainly of nominal payments, their value will behave like a short position in a long-term bond.

$$\text{Surplus} = \text{Assets} - \text{Liabilities}$$

$$\Delta \text{Surplus} = \Delta \text{Assets} - \Delta \text{Liabilities} = A \cdot R_A - L \cdot R_L$$

$$\begin{aligned} R_{\text{surplus}} &= \frac{\Delta \text{Surplus}}{\text{Asset}} = \frac{\Delta \text{Asset}}{\text{Asset}} - \frac{\Delta \text{Liabilities}}{\text{Liabilities}} \times \frac{\text{Liabilities}}{\text{Asset}} \\ &= R_{\text{asset}} - R_{\text{liability}} \frac{L}{A} \end{aligned}$$

- Example (Δ surplus):
 - Neptune employee retirement fund had \$900 million in assets and \$800 million in liabilities. At the end of 2007, due to the huge crash, the investment in asset lost 50% of their market value. The liabilities of the fund can be thought as a long-term debt with modified duration of 18. The interest rate declined 1.5%. What was the surplus before the huge crisis and what was the change in the pension fund's surplus?
- Answer:

$$\text{Surplus} = \text{Assets} - \text{Liabilities} = 900 - 800 = 100$$

$$\begin{aligned} \Delta \text{Surplus} &= \Delta \text{Assets} - \Delta \text{Liabilities} \\ &= 900 \times (-50\%) - (-\text{ModDur} \times \Delta y \times P) \\ &= 900 \times (-50\%) - [-18 \times (-1.5\%) \times 800] = -666 \end{aligned}$$

6 Funding risk - SaR

- Surplus at risk (SaR):

$$\sigma_{\text{surplus}} = \sqrt{A^2 \cdot \sigma_A^2 + L^2 \cdot \sigma_L^2 - 2 \cdot AL \sigma_A \sigma_L \rho} \quad (\text{dollar term}) \text{SaR} = |E(\Delta \text{Surplus}) - Zc \cdot \sigma_{\text{surplus}}|$$

- Example (SaR): Neptune employee retirement fund had \$900 million in assets and \$800 million in liabilities:

	Asset	Liabilities
Return	9%	6%
Volatility	15%	20%

Assume that the correlation between asset return and liability growth is 0.5. Please compute 95% surplus at risk.

- Answer:

$$\begin{aligned}
 \text{Expected surplus growth} &= A \cdot R_A - L \cdot R_L \\
 &= 900 \cdot 9\% - 800 \cdot 6\% = 33 \text{ million} \\
 \sigma_{\text{surplus}} &= \sqrt{A^2 \cdot \sigma_A^2 + L^2 \cdot \sigma_L^2 - 2 \cdot AL \sigma_A \sigma_L \rho} \\
 &= \sqrt{900^2 \cdot 0.15^2 + 800^2 \cdot 0.20^2 - 2 \cdot 900 \cdot 800 \cdot 0.15 \cdot 0.2 \cdot 0.5} \\
 &= 149 \\
 95\%SaR &= |33 - 1.645 \cdot 149| = 212.11 \text{ million}
 \end{aligned}$$

7 Funding risk - Confidence interval

- The confidence interval for expected end-of-year surplus

$$\begin{aligned}
 &[E(\text{Surplus}_1) - z \times \sigma_s, E(\text{Surplus}_1) + z \times \sigma_s] \\
 &E(\text{Surplus}_1) = \text{Surplus}_0 + E(\Delta \text{surplus})
 \end{aligned}$$

- Example (Confidence interval): information to the advisor or a pension fund that currently invests in government and corporate bonds:

Pension	Assets	Liabilities
Amount (EUR million)	200	180
Expected annual growth rate	5%	7%

Assuming that the volatility of surplus in dollars is EUR 36 million, what is the lower bound of the 95% confidence interval for the expected end-of-year surplus?

- Answer:
 - $\text{Surplus}_0 = 200 - 180 = 20 \text{ million}$
 - $\Delta \text{surplus} = 200 \times 5\% - 180 \times 7\% = -2.6 \text{ million}$
 - $E(\text{Surplus}) = \text{Surplus}_1 = \text{Surplus}_0 + \Delta \text{surplus} = 20 - 2.6 = 17.4 \text{ million}$
 - Lower bound of confidence interval: $E(\text{Surplus}) - z \times \sigma_s = 17.4 - 1.96 \times 36 = -53.16 \text{ million}$

8 Funding risk

- Sponsor risk: surplus risk can be extended to the risk to the owner of the fund who ultimately bear responsibility for the pension fund.
 - Economic risk: is the variation in the total economic earnings of the plan sponsor. (if the firm enjoys greater operating profits, the surplus risk may be less of a concern.)
 - Cash-flow risk: is the risk of year-to-year fluctuations in contributions to the pension fund.

9 VaR applications to investment management

- Sell side versus buy side:

Characteristic	Sell Side (Banks)	Buy Side (Investment Management)
Horizon	Short-term (1 day, intraday)	Long-term (month, quarter, years)
Turnover	Rapid	Slow
Leverage	High	Low
Risk measures	VaR, Stress test	Asset allocation, Tracking error
Risk controls	Position limits, VaR limits, Stop-loss rules	Diversification, Benchmarking, Investment guidelines

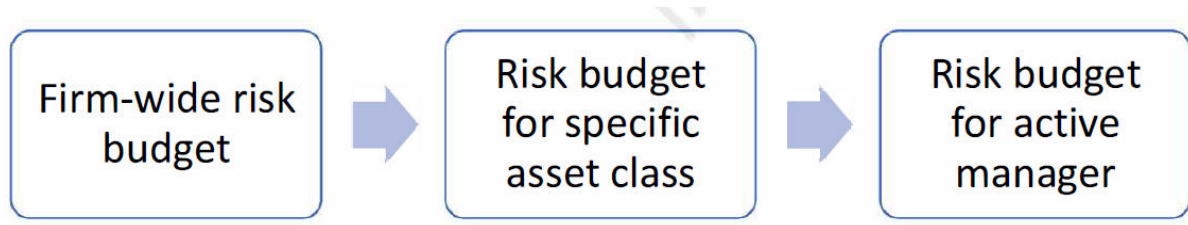
- Investment process of large investors (e.g. pension funds)
 - Step One: strategic, long-term asset-allocation study
 - * Based on mean-variance portfolio optimization
 - * Determines the amounts to be invested in various asset classes
 - Step Two: delegate fund managers
 - * Conduct performance evaluation periodically relative to their benchmark
 - * Investment guidelines and some additional restrictions: duration, maximum deviations from equity-sector weights, maximum amounts of foreign currency to hedge, etc.
 - Institutions exposed to a diversity of risk, to complex financial instruments, and to changing positions should benefit from VaR risk management systems.
 - * Take diversification into account.
 - * A need for stronger, centralized risk management systems (Financial instruments are becoming more complex).
 - * Most investment portfolios are dynamic.
- Hedge fund: poses special risk measurement problems.
 - High leverage
 - Lack of transparency
 - Heterogeneous
 - * Some groups have greater turnover than traditional investment managers → risk management systems similar to the bank

10 Using VaR to monitor and control risks

- Check compliance
 - Unauthorized investment: some securities may be prohibited because of their risks or for other reasons (e.g., political or religious). But some fund managers sometimes trade in and out of unauthorized investment before the client realized what happened.
- Monitor risk
 - Fund managers and investors would notice a sudden jump in the reported VaR of the fund. The reason may be:
 - * A manager taking more risk.
 - * Different managers taking similar bets.
 - * More volatile markets.
 - VaR can be reverse engineered to understand where risk is coming from using VaR tools (e.g., component VaR, marginal VaR).

11 Risk budgeting

- Risk budgeting: the process of decomposing the aggregate risk of a portfolio into its constituents, using these risk measures to allocate assets, setting limits in terms of these measures, and then using the limits to monitor the asset allocations and portfolio managers



12 Risk budgeting - across asset classes

- Example 1: a firm's management might set a annual return volatility target equal to 10%. If the firm has 1000 million in assets under management and assuming the returns are normally distributed, at a 99% confidence level.

table 4: Risk budgeting across asset classes at 99% (millions)

Asset	Weight	Volatility	Principal	Risk Budget
U.S. stocks	52.5%	15.1%	\$525	\$184
Non-U.S. stocks	10.4%	16.7%	\$104	\$41
U.S. bonds	12.25%	7.3%	\$122	\$21
Non-U.S. bonds	24.9%	10.9%	\$249	\$63
Portfolio	100%	10.0%	\$1000	\$233

$$VaR_{US\text{ stocks}} = 2.326 \cdot 0.151 \cdot 525 = 184$$

$$\text{Undiversified } VaR = \sum \text{Risk Budget} = \$309 > \$233$$

- Example 2: the current portfolio fund manager Henry holding is a €100 million investment with 10% volatility. The risk budget he set is €40million at 99% confidence level. Which investment should he add into current portfolio can keep his risk budget?

	Return volatility	Correlation with current portfolio	Value
Emerging market stocks	15%	-0.14	100million
Investment grade bonds	8.5%	0.8	100million

$$VaR_{\text{stock}} = 15\% \cdot 100 \cdot 2.326 = 34.89\text{million}$$

$$VaR_{\text{bond}} = 8.5\% \cdot 100 \cdot 2.326 = 19.77\text{million}$$

$$VaR_{\text{current}} = 10\% \cdot 100 \cdot 2.326 = 23.26\text{million}$$

If adding emerging market stocks into portfolio:

$$VaR_{\text{new}} = \sqrt{34.89^2 + 23.26^2 - 2 \cdot 34.89 \cdot 23.26 \cdot 0.14} = 39.13\text{million}$$

If adding investment grade bonds into portfolio:

$$VaR_{\text{new}} = \sqrt{19.77^2 + 23.26^2 + 2 \cdot 19.77 \cdot 23.26 \cdot 0.8} = 40.84 \text{million}$$

13 Risk budgeting - across active managers

- A greater risk budget should be allocated to managers with better performance, as measured by the IR criterion.
- Weight of portfolio managed by manager i:

$$w_i = \frac{\frac{IR_i}{TE_i}}{\frac{IR_p}{TE_p}} = \frac{\frac{\alpha_i}{TE_i^2}}{\frac{\alpha_p}{TE_p^2}}$$

Assume that the deviations for each manager are independent of each other

TE: track error

- Example: PERF wants to allocate \$525 million to a pool of active managers so as to maximize the information ratio of the fund subject to an overall tracking error (TE) of 4%.
- Please calculate the weight of portfolio and relative risk budget which is allocated to manager 1.

	TE_i	IR_i
Manager 1	6%	0.6
Manager 2	6%	0.4
Portfolio	4%	0.72

- Answer:

table 5: Risk Budgeting for Portfolio Managers

	TE_i	IR_i	Weight	Allocated Principal	Relative risk budget (99% confidence level)
Manager 1	6%	0.6	55.6%	\$291	\$40.6
Manager 2	6%	0.4	37.04%	\$194	\$27.1
Index	0%	0	7.36%	\$40	\$0
Portfolio	4%	0.72	100%	\$525	\$48.85

$$Weight_{\text{manager 1}} = \frac{\frac{IR_i}{TE_i}}{\frac{IR_p}{TE_p}} = \frac{\frac{0.6}{6\%}}{\frac{0.72}{4\%}} = 55.6\% \quad (5)$$

$$\text{Amount} = 55.6\% \times 525 \text{million} = 291.9 \text{million} \quad (6)$$

$$\text{Relative risk budget} = z \cdot \sigma_\alpha \cdot V = 2.326 \times 291 \times 6\% = 40.61 \text{million}$$

- The residual weight is allocated to the index.

考点 4 Risk Monitoring and Performance Measurement

1 VaR vs. Tracking error

- VaR: the maximum dollar loss associated with a given level of statistical confidence over a given period of time.

- To achieve targeted levels of dollar VaR, owners of capital allocate capital among asset classes.
- Tracking error: gauges the risk profile relative to a benchmark (standard deviation of excess returns).
 - Tracking error is used to describe the extent to which the investment manager is allowed latitude to differ from the index.

2 Three-legged risk management stool



Quantify the vision of the plan

- Risk plan
 - Set expected return and expected volatility (e.g., VaR and tracking error)
 - Define the bright line between those events that are merely disappointing and those that inflict serious damage.
 - Paint a vision of how risk capital will be deployed to meet the organization's objectives.
 - The risk plan should identify critical dependencies that exist inside and outside the organization.
- Risk budget
 - The risk budget should quantify the vision of the risk plan
 - * Using mean-variance optimization to determine appropriate weights for each class (expected returns, risks, and covariances, organization's strategic objectives and risk tolerances).
 - * Ensure ROE and RORC must exceed some minimum levels.
 - * Simulate the performance of a portfolio (e.g. downside scenarios)

- Risk monitoring
 - Risk capital is a scarce commodity, so monitoring controls should exist to ensure that risk capital is used in a manner consistent with the risk budget.
 - * Risk monitoring is required to ensure that material deviations from risk budget are detected and addressed in a timely fashion.

3 Risk management units (RMUs)

- RMUs:
 - Role: oversee the risk exposures of portfolios and ensure that such exposures are authorized and in line with risk budgets.
 - Must be established independently from the business areas and operate as a controlling or monitoring function.
 - * have an independent reporting line to senior management.
 - Gathers, monitors, analyzes, and distributes risk data to managers, clients, and senior management in order to better understand and control risk.
 - Should not manage risk, but rather measure risk.
 - Watches trends in risk as they occur and identifies unusual events to management in a timely fashion.
 - Identifies and develops risk measurement and performance attribution analytical tools
 - Provides tools for both senior management and individual portfolio management to better understand risk in individual portfolios and the source of performance.
 - Helps ensure that transactions are authorized in accordance with management direction and client expectations.

4 Liquidity Considerations

- Liquidity duration: the number of days required to liquidate any given security.

$$LD_i = \frac{Q_i}{0.15 \cdot V_i}$$

LD: liquidity duration statistic for security

15%: we do not wish to exceed 15% of the daily volume in that security to avoid a **material adverse earning impact**.

Q: number of shares held in security

V: daily volume of the security

5 Performance measurement

- Objectives of performance measurement tools:
 - To determine whether a manager generates consistent excess risk-adjusted performance to a benchmark.
 - To determine whether a manager generates superior riskadjusted performance to the peer group.
 - To determine whether the returns achieved are sufficient to compensate for the risk assumed in cost/ benefit terms.
 - To identifying those managers whose processes generate high-quality excess risk-adjusted returns.
- Tool 1: The green zone - For prior week, month, year: calculate normalized returns (excess returns) and tracking error. Then, Compare actual outcomes to target outcomes.
- Policy decisions about deviations:
 - Green zone: An acceptable amount of deviation.
 - Yellow zone: Unusual events that are expected to occur with some regularity.

- Red zone: Truly unusual events and requiring immediate follow-up.
- Tool 2 - Attribution of returns
- Tool 3 - The Sharpe and information ratios
- Tool 4 - Alpha versus the benchmark
- Tool 5 - Alpha versus the peer group

考点 5 Portfolio Performance Evaluation

1 Return calculation

- Time-weighted returns: The compound return that \$1 initially invested in the portfolio over a stated measurement period.

$$(1 + R_G)^n = (1 + R_1)(1 + R_2)(1 + R_3) \cdots (1 + R_n)$$

$$1 + R_G = [(1 + R_1)(1 + R_2)(1 + R_3) \cdots (1 + R_n)]^{1/n}$$

- Each return has an equal weight in the geometric average. The geometric average is referred to as a time-weighted average.
- Dollar-weighted returns: accounts for the timing and amount of all cash flows into and out of the portfolio.
- Calculation of DWR: similar to IRR.

$$CF_0 + \frac{CF_1}{1 + DWR} + \cdots + \frac{CF_n}{(1 + DWR)^n} = 0$$

- Example:

table 6: Stock Investment Cash Flows

Time	Outlay / Proceeds
0	\$50 to purchase first share
1	\$53 to purchase second share a year later
1	\$2 dividend from initially purchased share
2	\$4 dividend from the 2 shares in the second year, plus \$108 received from selling both shares at \$54 each

TWR:

$$R1 = \frac{53 + 2 - 50}{50} = 10\%$$

$$R2 = \frac{108 + 4 - 53 \cdot 2}{53 \cdot 2} = 5.66\%$$

$$TWR = \sqrt{(1 + R1)(1 + R2)} - 1 = 7.8\%$$

DWR:

$$0 = -50 + \frac{-53 + 2}{1 + DWR} + \frac{108 + 4}{(1 + DWR)^2} \rightarrow DWR = 7.117\%$$

计算器上的 F 表示 frequency, 即这笔现金流出现了几次由于题目中现金流都是只出现一次, 默认为 1 往下翻即可。

- DWR: If more funds to invest:
 - Before favorable time: the dollar-weighted rate of return will tend to be elevated (higher/increased).
 - Before unfavorable time: the dollar-weighted rate of return will tend to be depressed (lower/decreased).
- TWR: The use of the time-weighted return removes distortions of fund investing, providing a better measure of a manager's ability to select investments over the period.

table 7: Calculation Steps for IRR

Steps	Display
[CF]	CF0=0.0000
[2nd][CE/C]	CF0=0.0000 (Clear previous works)
[50][+/-][ENTER]	CF0=-50.0000
[↓][51][+/-][ENTER]	C01=-51.0000
[↓][↓][112][ENTER]	C02=112
[IRR][CPT]	IRR=7.117%

2 Risk-adjusted performance measures

- Sharpe ratio: divides average portfolio excess return over the sample period by the standard deviation of returns (total risk) over that period.

$$\text{Sharpe Ratio} = \frac{\overline{R_p} - \overline{R_f}}{\sigma_p}$$

- Treynor ratio: like the Sharpe ratio, Treynor's measure gives excess return per unit of risk, but it uses systematic risk instead of total risk.

$$\text{Treynor Ratio} = \frac{\overline{R_p} - \overline{R_f}}{\beta_p}$$

- Sharpe ratio vs. Treynor ratio:
 - Well-diversified portfolio: performance evaluation using Sharpe ratio and Treynor ratio will give the same ranking result.
 - Not well-diversified portfolio: it may be ranked relatively higher using Treynor ratio than using Sharpe ratio.
- Jensen's alpha: is the average return on the portfolio over and above that predicted by the CAPM, given the portfolio's beta and the average market return.

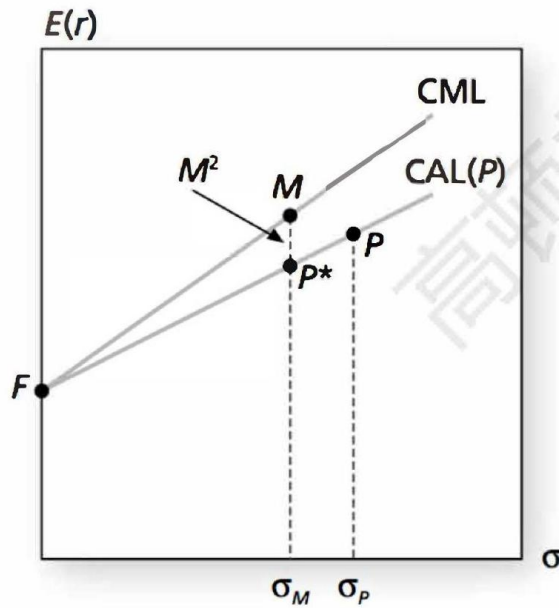
$$\alpha_p = \overline{r_p} - [\overline{r_f} + \beta_p(\overline{r_m} - \overline{r_f})]$$

- Information ratio: measures excess return over benchmark per unit of risk.

$$IR = \frac{\alpha_p}{\sigma(\alpha_p)}$$

- A higher information ratio indicates better performance.
- Modigliani-squared measure (m^2):
 - we adjust the return of portfolio to the same standard deviation as the index. Because the market index and portfolio have the same standard deviation, we can compare their performance simply by comparing returns.

$$M^2 = \frac{\sigma_m}{\sigma_p}(R_p - R_f) - (R_m - R_f)$$



- Example: consider the following data for a particular sample period

table 8: Portfolio and Market Metrics

	Portfolio P	Market M
Average return	35%	28%
Beta	1.2	1
Standard deviation	42%	30%
Tracking error	18%	0

- Please calculate Sharpe, Jensen (alpha), Treynor, information ratio and m^2 . The T-bill rate during the period was 6%. By which measures did portfolio outperform the market.
- Answer:

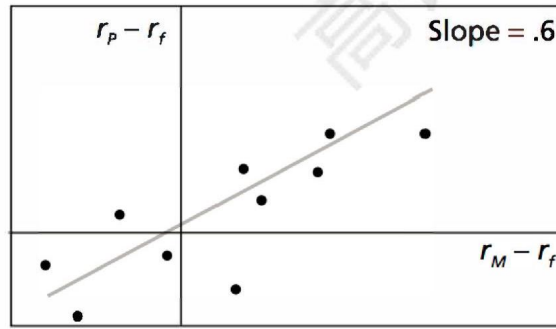
table 9: Performance Metrics Comparison

	Portfolio P	Market M
Sharpe ratio	0.6905	0.7333
Treynor ratio	0.2417	0.22
Jensen's alpha	0.026	0
Information ratio	0.3889	0
M^2	-0.0129	0

3 Market timing

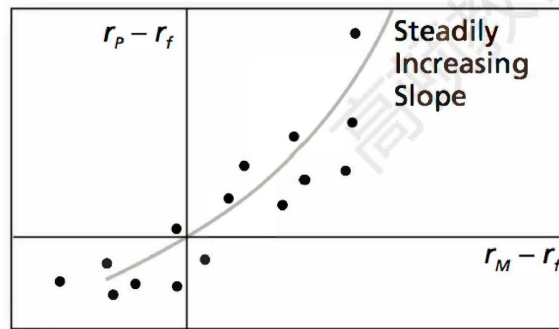
- Market timing involves shifting funds between a market-index portfolio and a safe asset depending on whether the market index is expected to outperform the safe asset.
- Assume fund manager holds only the market-index portfolio and T-bills.
- No market timing (the weight of the market are constant)

$$R_p - R_f = a + b(R_m - R_f) + e_p$$



- With market timing (Treynor and Mazuy)

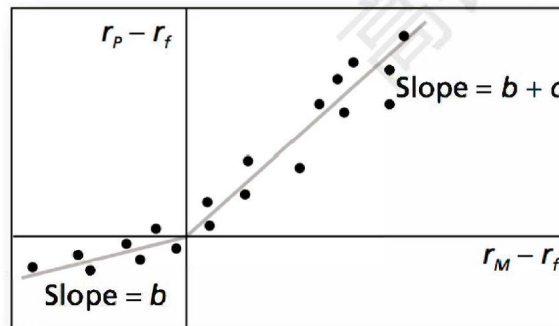
$$R_p - R_f = a + b(R_m - R_f) + c(R_m - R_f)^2 + e_p$$



- With market timing (Henriksson and Merton): the beta of the portfolio takes a large value if the market is expected to do well and a small value otherwise.

$$R_p - R_f = a + b(R_m - R_f) + c(R_m - R_f)D + e_p$$

- D is a dummy variable with 0 for bear markets ($R_M < R_f$) and 1 for bull markets ($R_M > R_f$).



- Call option model: The perfect foresight is equivalent to holding a call option on the equity portfolio. The rate of return is at least the risk-free rate.
- Suppose that the market index currently is at S_0 and that a call option on the index has an exercise price of $X = S_0 \square 1 + R_f \square$

table 10: Option Payoff Analysis

	$S_T < X$	$S_T \geq X$
Bills	$S_0(1 + R_f)$	$S_0(1 + R_f)$
Call	0	$S_T - X$
Total	$S_0(1 + R_f)$	S_T

4 Performance attribution

- The attribution method explains the difference in returns between a managed portfolio, P, and a selected benchmark portfolio, B, called the bogey.
- Example: The portfolio return over the month is 5.34%

table 11: Bogey Performance

	Benchmark weight	Return of index during month(%)
Equity (S&P500)	60%	5.81
Bonds (Barclays aggregate index)	30%	1.45
Cash (money market)	10%	0.48

$$\text{Alpha} = 5.34\% - (60\% \cdot 5.81\% + 30\% \cdot 1.45\% + 10\% \cdot 0.48\%) = 1.37\%$$

- Three components in attribution system:
 - Asset allocation: Broad asset market allocation choices across equity, fixed-income, and money markets.
 - Selection:
 - * Industry (sector) choice within each market.
 - * Security choice within each sector.
- Example (cont.):

table 12: Portfolio and Benchmark Performance

	Equity	Bonds	Cash
Return of index during month(%)	5.81	1.45	0.48
Benchmark weight	60%	30%	10%
Portfolio performance (%)	7.28	1.89	0.48
Actual weight	70%	7%	23%

$$\begin{aligned}
 &\text{Contribution from asset allocation} && (W_{Pi} - W_{Bi}) \times R_{Bi} \\
 &+ \text{Contribution from security selection (selection)} && (R_{Pi} - R_{Bi}) \times W_{Pi} \\
 &= \text{Total contribution from asset class i} && W_{Pi}R_{Pi} - W_{Bi}R_{Bi}
 \end{aligned}$$

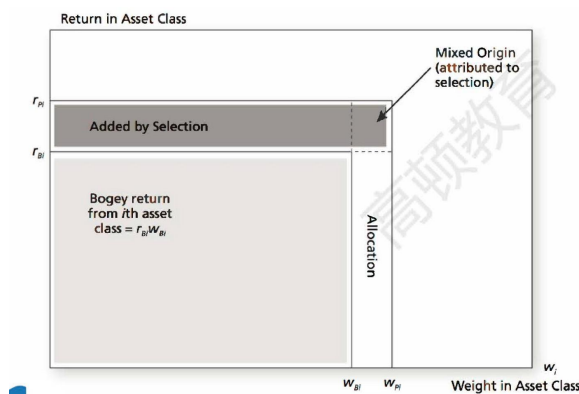


table 13: Portfolio Performance Analysis

	Actual weight	Bench weight	markActive weight	Portfolio perfor-mance(%)	Return of dex(%)	in-Excess perfor-mance(%)
Equity	70%	60%	10%	7.28	5.81	1.47
Bonds	7%	30%	-23%	1.89	1.45	0.44
Cash	23%	10%	13%	0.48	0.48	0

Asset allocation= $0.1 \cdot 5.81\% + (-0.23) \cdot 1.45\% + 0.13 \cdot 0.48\% = 0.003099$

Security selection= $0.7 \cdot 1.47\% + 0.07 \cdot 0.44\% = 0.0106$

- Example (cont.): sector selection

table 14: Sector - level Portfolio Performance Analysis

Sector	(1)Beginning of Month Weights (%)		(3)	(4)	(5) = (3) × (4)	
	Portfolio	S&P 500			Sector	Allocation Contribution
Basic materials	1.96	8.3	-6.34	6.9	-0.4375	
Business services	7.84	4.1	3.74	7.0	0.2618	
Capital goods	1.87	7.8	-5.93	4.1	-0.2431	
Consumer cyclical	8.47	12.5	-4.03	8.8	0.3546	
Consumer noncyclical	40.37	20.4	19.97	10.0	1.9970	
Credit sensitive	24.01	21.8	2.21	5.0	0.1105	
Energy	13.53	14.2	-0.67	2.6	-0.0174	
Technology	1.95	10.9	-8.95	0.3	-0.0269	
TOTAL					1.2898	

table 15: Portfolio Performance Decomposition

	(1) Actual weight	(2) Benchmark weight	(3) Active weight (1)-(2)	(4) Portfolio perfor-mance	(5) Return of index	(6) Excess perfor-mance (4)-(5)
Equity	0.7	0.6	0.1	7.28%	5.81%	1.47%
Bonds	0.07	0.3	-0.23	1.89%	1.45%	0.44%
Cash	0.23	0.1	0.13	0.48%	0.48%	0%
Asset allocation					$\sum (3) \times (5)$	0.3099%
Selection					$\sum (1) \times (6)$	1.060%
Equity			1.47%	1.47%*0.7	1.0290%	
		-Sector	1.29%			
		-Security	0.18%			
Bond			0.44%	0.44%*0.07	0.0308%	
Total alpha						1.37%

3 Hedge funds

考点 1 Hedge Fund 对冲基金

1 Characteristics of hedge funds

table 16: Comparison between Hedge Funds and Mutual Funds

	Hedge Funds	Mutual Funds
Characteristic	Private vehicles	General Public
Regulation	Less Regulated	Severe Regulation
Leverage	High	Low
Strategy	Both long and short positions	Long - bias
Transparency	Low (keep from mimicking or front running)	High
Fees Charge	2 and 20, 1 and 10	Only Management Fees
Performance evaluation	Expect out - sized return	Well defined performance target

2 Databases of hedge funds

- Self-Selection bias (selection bias): Manager who have funds with an unimpressive track record will not wish to have that information exposed. In other words, there may well be a tendency to "put the best face forward".
- Backfill bias: When a new fund replaces a deleted fund in an index, the past performance of the new fund is inserted.
- Issues from illiquid assets and the solutions
 - Correlation with other asset classes will be lowered (appearance of low systematic risk)
 - * Using regression with additional lags of the market factors and summing the coefficient across lags.
 - Volatility will be lowered (low total risk)
 - * Taking autocorrelation into account when extrapolating risk to longer horizons.

3 Evolution of hedge fund industry

- Early days (1987-1994): show a very sizeable performance gap between the large hedge funds and the SNP.
- Fall of LTCM (1998): a significant reduction in investors' appetite.
- Burst of the dot-com bubble (2000-2001)
- The arrival of institutional investors (2002-2010): Investors in hedge funds thus shifted from exclusively private wealth to institutions, including foundations, endowments, pension funds, and insurance companies.
 - They entered hedge fund market in search of alternative sources of return away from the equity market.

4 Hedge fund strategies

- Directional hedge fund styles:
 - Managed futures
 - Global Macro
- Event-driven styles:
 - Merger arbitrage
 - Distressed securities

- Relative value and arbitrage-like strategies:
 - Fixed income arbitrage
 - Convertible arbitrage
 - Long/short equity
- Niche strategies:
 - Dedicated short
 - Emerging markets
 - Equity market neutral strategy
- Directional hedge fund styles:
 - Managed futures (CTAs, commodity trading advisors) : focus on investing in listed bond, equity, commodity futures, and currency markets, globally.
 - * Systematic trading programs are employed. Managed futures fund managers rely upon historical price data and market trends.
 - * A significant amount of leverage is often employed since the strategy involves the use of futures contracts.
 - Global macro:
 - * Focuses on identifying extreme price valuations and leverage is often applied on the anticipated price movements in equity, currency, interest rate and commodity markets.
 - * employs a top-down global approach to concentrated on forecasting how political trends and global macroeconomic events affect the valuation of financial instruments.
 - * Profits can be made by correctly anticipating price movements in global markets and having the flexibility to use a broad investment mandate, with the ability to hold positions in practically any market with any instrument.
 - Similarity between global macro and managed futures: these managers behave like asset allocators taking bets in different markets utilizing a range of strategies opportunistically.
- Event-driven styles:
 - Merger arbitrage: attempts to capture the spreads in merger or acquisition transactions involving public companies after the terms of the transaction have been announced.
 - In a fixed exchange ratio stock merger, one would go long one and short another one according to the merger ratio, in order to isolate the spread and hedge out market risk.
 - Distressed securities: This strategy invests across the capital structure of companies subject to financial or operational distress or bankruptcy proceedings. Such securities often trade at discount to intrinsic value due to difficulties in assessing their proper value, lack of research coverage, or an inability of traditional investors to continue holding them.
 - * Distressed managers typically attempt to profit on the issuer's ability to improve its operation or the success of the bankruptcy process that ultimately leads to an exit strategy.
 - * The main feature of this investment strategy is long exposure to credit risk of corporations with very low credit ratings.
 - This strategy is generally long-biased in nature.
 - There is some evidence of nonlinearity between the two return series, particularly at the extreme tails (mostly as tail risk that shows up under extreme market conditions).
 - Unlike trend followers, who tend to benefit from extreme moves, event-driven funds are hurt by extreme moves.
- Relative value and arbitrage-like strategies:

- Fixed income arbitrage: attempts to generate profits by exploiting inefficiencies and price anomalies between related fixed income securities.

table 17: Financial Instrument Strategies

Instrument	Strategy (Bet)
Swap spread trade	Change in spread of fixed side and floating side
Yield - curve spread trades	Deviate from overall yield curve only for short - run
Mortgage spread trades	Prepayment rates
Fixed income volatility trades (Gamma, Vega)	Implied volatility of interest
Capital structure or credit arbitrage trades	Mispricing among different types of securities (equity and debt)

- Convertible arbitrage: aims to profit from the purchase of convertible securities and the subsequent shorting of the corresponding stock when there is a pricing error.
 - * The number of shares shorted is based on a delta neutral or market neutral ratio. As a result, under normal market conditions, the arbitrageur generally expects the combined position to be insensitive to fluctuations in the price of the underlying stock.
 - * Pricing anomaly
 - * Gamma/vega trading
- Long/short equity: invests in both long and short sides of equity markets, generally focusing on diversifying or hedging across particular sectors, regions or market capitalizations.
 - * Managers typically have the flexibility to shift from value to growth; small to medium to large capitalization stocks; and net long to net short.
 - * 30-40% of hedge funds are long/short.
- Niche strategies:
 - Dedicated short: takes more short positions than long positions and earn returns by maintaining net short exposure in long and short equities.
 - * The returns are negatively correlated to equities.
 - Emerging markets: invests in currencies, debt, equities, and other instruments of countries with emerging or developing markets.
 - * The markets are typically measured by GDP per capita. Such countries are considered to be in a transitional phase between developing and developed status.
 - Equity market neutral strategy: there is no single common risk factor that drives the return behavior of equity market neutral funds. Return behavior suggests that different funds apply different trading strategies with a similar goal of achieving almost zero beta(s) against a broad set of equity indices.
- Exercise
 - Samantha Moore manages a hedge fund for a mid-sized money management firm. The fund frequently changes styles according to identified profit opportunities. At the beginning of the year, the fund took a long position in 10-year subordinated 8% coupon debt issued by a firm expected to undergo reorganization under Chapter 11. Moore felt that analysts had been paying too little attention to the issuer. Six months later, the fund completed a second transaction involving a long position in Swiss Francs and a short position in Japanese Yen based on forecasted movements in interest rates in the two countries. What two hedge fund strategies are most likely being employed by Moore's hedge fund?

- A. Distressed securities strategy and equity long/short strategy.
- B. Fixed-income arbitrage and global macro strategy.
- C. Distressed securities strategy and global macro strategy.
- D. Fixed-income arbitrage and equity long/short strategy.

– Answer: C

解析: 基金购入了一只次级债, 公司正在进行破产充足, 此为破产公司证券策略: 六个月后, 基金进入了瑞士法郎的多头和日元的空头, 主要是在宏观上对这两个国家的货币做出相对的选择, 此为全球宏观策略。

5 Hedge fund performance

- In practice, evaluating hedge funds poses considerable practical challenges:
 - The risk profile of hedge funds may change rapidly.
 - * Take time-varying bets on different risk factors and the risk profile is nonlinear and responsive to changing equity market conditions.
 - * Considerable ability to manipulate conventional performance measures.
 - * Expose the fund to infrequent but severe losses.
 - Hedge funds tend to invest in illiquid assets.
 - When hedge funds are evaluated as a group, survivorship bias can be a major consideration, because turnover in this industry is higher than for investment companies such as mutual funds.

6 Convergence of risk factors

- An important feature that attracts investors to the hedge fund industry is the variety of strategies hedge fund managers deploy and the diversity of assets to which these strategies are applied.
- However, during times of stress, portfolio diversification implodes, and seemingly diverse hedge fund portfolios converge in terms of risk factors.
- March to April period in 1994, unexpected hike in interest rate policy by the US Federal Reserve.
- August 1998 leading up to the collapse of LTCM.
- 2007-2008 financial crisis: "forced liquidation" of leveraged positions driven by a combination of rising margin requirements, borrowing costs and investors withdrawing their equity.
- The risk behind cannot be easily mitigated by simply spreading one's capital to different hedge fund strategies.
- The need to consider hedging a tail risk event in a hedge fund portfolio is clear.
- The risk management of hedge fund portfolios must take into consideration nonlinear risks such as rare but dramatic market events the manifestation of which can take place on either the asset side and/or the liability side of a hedge fund's balance sheet.

7 Agency problem

- There is a concern about the asymmetry in risk sharing between principal and agent inherent in variable compensation schemes of which hedge fund style incentive fees is one such variation.
- The problem arises when the incentive fee which hedge fund managers are entitled to, typically at 15-20% of new profits, ends up enticing a fund manager to take unreasonable bets.
 - If fund has a negative return, managers still obtain management fee.
- How to reduce asymmetric risk:

- High water mark.
- Lower incentive fees for fund managers.
- High fund closure costs.
- Force a fund manager to invest in any fund that he manages.

考点 2 Performing Due Diligence on Specific Managers and Funds

1 Due diligence basics

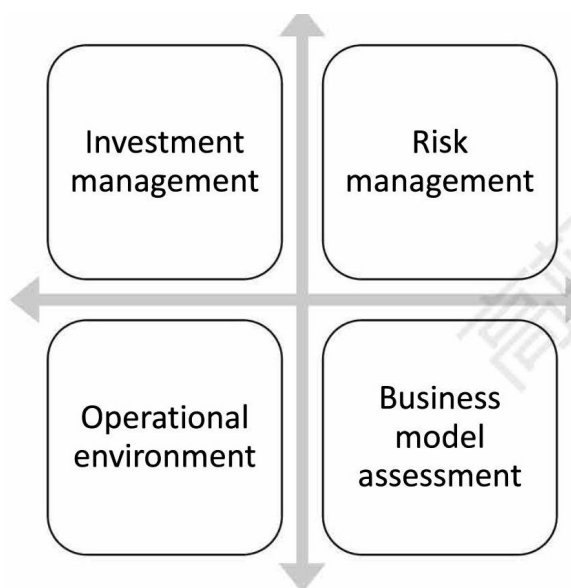
- Due diligence is the term used to describe the process of evaluation and analysis that an investor follows to get comfortable with a strategy, manager, and a fund prior to making an investment.

2 Reasons for failures of funds in the past

- Bad investment decisions: funds can make several compounded bad decisions or just a few concentrated calls on the markets or individual securities that perform very poorly.
- Fraud: accounting frauds, valuation frauds, or misappropriation of funds.
- Extreme events: excessive leverage, improbable probabilities, unexpected events and tail risk.
- Lack of liquidity:
 - A flood of unanticipated withdrawals of capital at the least opportune time.
 - Squeeze.
 - When liquidity dries up, funds cannot meet redemptions.
- Poor controls: a lack of supervision or compliance controls related to insider trading.

3 Elements of the due diligence process

- Today, both managers and investors spend a great deal of time trying to learn where a manager's "edge" is coming from and that their investment is safeguarded and properly valued.
- The goal is to find out what is really going on at the fund and not just have a pitch book recitation by the investor relations staff.



4 Investment management

- Strategy:
 - Manager's self-described style
 - Concentrated positions
 - Turnover and number of days to liquidate
 - Stop losses
 - Short sales used
 - OTC derivatives
 - Central trading desk
 - Generate returns for existing investors or grow AUM?
- Equity ownership: some firms do not share equity ownership among the portfolio managers or traders in the firm; other firms use equity ownership as a key feature of their professional talent retention program and to attract and groom new talent for future leadership position.
- Track record reliable
 - Whether the track record is comparable to similar strategies
- Managers:
 - Newer managers: ask former colleagues, partners, managers, clients, or other independent parties.
 - Established managers: motivation, behavior during crisis or stress periods; ability to communicate to investor.

5 Risk management evaluation

- Risk (Assessment, measurement, monitoring, mitigating)
- Potential risks that a strategy will be exposed and additional idiosyncratic risks
- Inputs and assumptions used in the firm's risk models
- Written policies and procedures to monitor and measure each of the applicable risk
- Security valuation
- Portfolio leverage and liquidity
- Tail risk exposure (Return distribution, skewness or kurtosis.)
- Risk reports (Review and revise on a regular basis)
- Consistency of the fund terms with the investment strategy.

6 Operational environment

- Internal control assessment
 - Qualifications of people at the fund
 - The quality of the written policies and procedures
 - The ability of the team to execute, fund's exposure to derivative counterparties
 - The protections provided by its governance structure
 - In-house compliance or outsourced relationship with a compliance service provider
- Documents and disclosure

- An investor needs to verify with the listed law firm in the fund documents that they were responsible for the original content and for any updates.
- Fund offering memo, subscription agreement, limited partnership agreement, investment management agreement, Form ADV and website are all saying the same thing at the same point in time.
- Either insufficient or extremely broad risk disclosure
- Check the equity section to see if the general partner is continuing to invest in the fund
- Service provider evaluation

7 Business model risk and fraud risk

- Business model risk: deals with the risks encompassed in simply running the business of being a hedge fund.
 - Adequate cash on hand.
 - Where does the working capital come from.
 - Succession plan (key person insurance).
 - What happens if too many investors redeem at the same time.
 - Break-even AUM.
- Fraud risk
- Potential indicators:
 - Lack of trading independence.
 - Investor complaints about lack of liquidity.
 - Litigation in civil court alleging fraudulent acts.
 - Personal trading by managers
 - Aggressive shorting and organized efforts to spread rumors.
 - A high percentage of illiquid investment

8 Due diligence questionnaire

- 1. Manager information
 - a. Registration
 - b. Ownership
 - c. Organization
 - d. References/ Background checks
 - e. Track record
 - f. Risk management
 - g. Operations
 - h. Service providers
 - i. Contact details
- 2. Execution and trading
- 3. Third-party research policy
- 4. Compliance
- 5. In-house legal

- 6. Anti-money-laundering policy and procedures
- 7. Disaster recovery and business continuity
- 8. Insurance coverage and key man provisions
- 9. Fund information
 - a. Management and incentive fees
 - b. Lock-up
 - c. Subscriptions and redemptions
 - d. Notice periods
 - e. Fund directors or advisors
 - g. Auditor
 - h. Legal advisor
 - k. Performance
 - l. Capacity
 - m. Gates and lock-ups
 - n. Historical drawdown
 - o. Use of managed accounts
 - p. Investor mix
- 10. Investment process and portfolio construction
- 11. Risk controls
 - a. Concentration and diversification
 - b. Liquidity
 - c. Leverage
 - d. Exposure to market risk factors
 - e. Reporting
 - f. Portfolio Greeks (duration, delta, beta, etc.)
- 12. Financial statements
 - a. Year-end opinion
 - b. Level 1, 2, 3 assets
 - c. Interim statements
 - d. Administrator reports
- 13. Terms and use of third-party marketers

考点 3 Finding Bernie Madoff: Detecting Fraud by Investment Managers

1 Fraud by investment managers

- Ponzi scheme: on December 11, 2008, the securities and exchange commission (SEC) charged Bernard Madoff with securities fraud for committing an \$18 billion Ponzi scheme.
 - The opportunities advisers have to exploit investors.
 - The importance of limiting advisers' opportunistic behavior through either market or regulatory forces.
 - In the U.S., the regulatory system protects investors primarily through mandatory disclosure.

- Fraud: fraud cases harm the firm's investment clients.
 - Not include insider trading, short sale violations, brokerage fraud, or other crimes, unless they cause direct losses to the firm's investment clients.

2 Information disclosure

- Form ADV:
 - Investment advisers: any entity that receives compensation for managing securities portfolios or providing advice regarding individual securities.
 - * Firms: all mutual funds, nearly all institutional investment funds, and many hedge funds in the U.S.
 - Registered investment advisers: must file Form ADV to disclose past regulatory violations and potential conflicts of interest.

table 18: Disclosure Items

Items 1 - 6	Items 7 - 8	Item 9	Item 10	Item 11	Item 12
Descriptive information about a firm and its operations	Disclosure of certain conflicts of interest	Disclosure regarding the custody of clients' assets	Disclosure of control person	Disclosure of past legal and regulatory violations	Small business
Schedules A - C: direct and indirect owners of a firm					
Schedule D: disclosure of affiliations with other financial firms					

- An investor who avoided the 5% of firms with the highest ex ante predicted fraud risk would avoid 29% of fraud cases and over 40% of the dollar losses from fraud.
 - Although to obtain these benefits, the investor would have to forgo investing with 5% of non-fraudulent advisers.

3 Predicting fraud

Disciplinary history and regulatory and legal offenses	Past fraud	×
	Past affiliated fraud	×
	Past regulatory	✓
	Past civil or criminal	✓
Conflicts of interest	Referral fees	✓
	Interest in transaction	✓
	Soft dollars	×
Monitoring	Broker in firm	✓
	Investment Company Act	✓
	Custody	×
	Dedicated CCO	×
Client Type	Hedge fund clients	×

table 19: Compliance and Monitoring Overview

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4 Compensated for fraud risk

- Are investors compensated for fraud risk?
 - High fraud risk may provide off-setting benefits that improve investment performance.
- Results: do not provide evidence that investors receive compensation for fraud risk.

5 Barriers and costs

- The lack of compensation for fraud risk does not necessarily imply that investors are irrational.
 - Investors could be compensated in some other unmeasured fashion.
 - Investors may predict fraud risk based on reputations, personal contacts, or other information
 - Barriers to accessing Form ADV outweigh the benefits.
- For most investors, the cost of individually downloading thousands of Form ADV filings may well have exceeded the perceived benefits.
 - To implement a fraud prediction model, an investor would have had to collect manually a large number of Form ADV filings, convert the filings into a database, and estimate a prediction model.

6 Improving ways

- Allowing public access to historical Form ADV filings would reduce the marginal cost of increased enforcement by facilitating investors' use of these data.
- This should reduce the marginal benefit to an investment adviser of committing fraud due to an increase in the probability of detection.
- Improved public access to these disclosure data could reduce the occurrence of fraud.